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Course code:DSA0602

Class Lab Practical

Set 1: Monthly Sales Data

Dataset: Monthly Sales

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.
2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.
3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.
4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

Python Program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read your CSV file
```

```
data = pd.read_csv("Book1.csv")
```

```
# Line Chart
```

```
plt.plot(data["Month"], data["Sales (in $)"], marker="o")
```

```
plt.title("Monthly Sales")
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Sales ($)")
```

```
plt.show()
```

```
# Bar Chart
```

```
plt.bar(data["Product"], data["Product Sales"])
```

```
plt.title("Top Selling Products")
```

```
plt.xlabel("Product")
```

```
plt.ylabel("Sales")
```

```
plt.show()
```

```
# Scatter Plot
```

```
plt.scatter(data["Advertising Budget"], data["Sales (in $)"])
```

```
plt.title("Advertising Budget vs Monthly Sales")
```

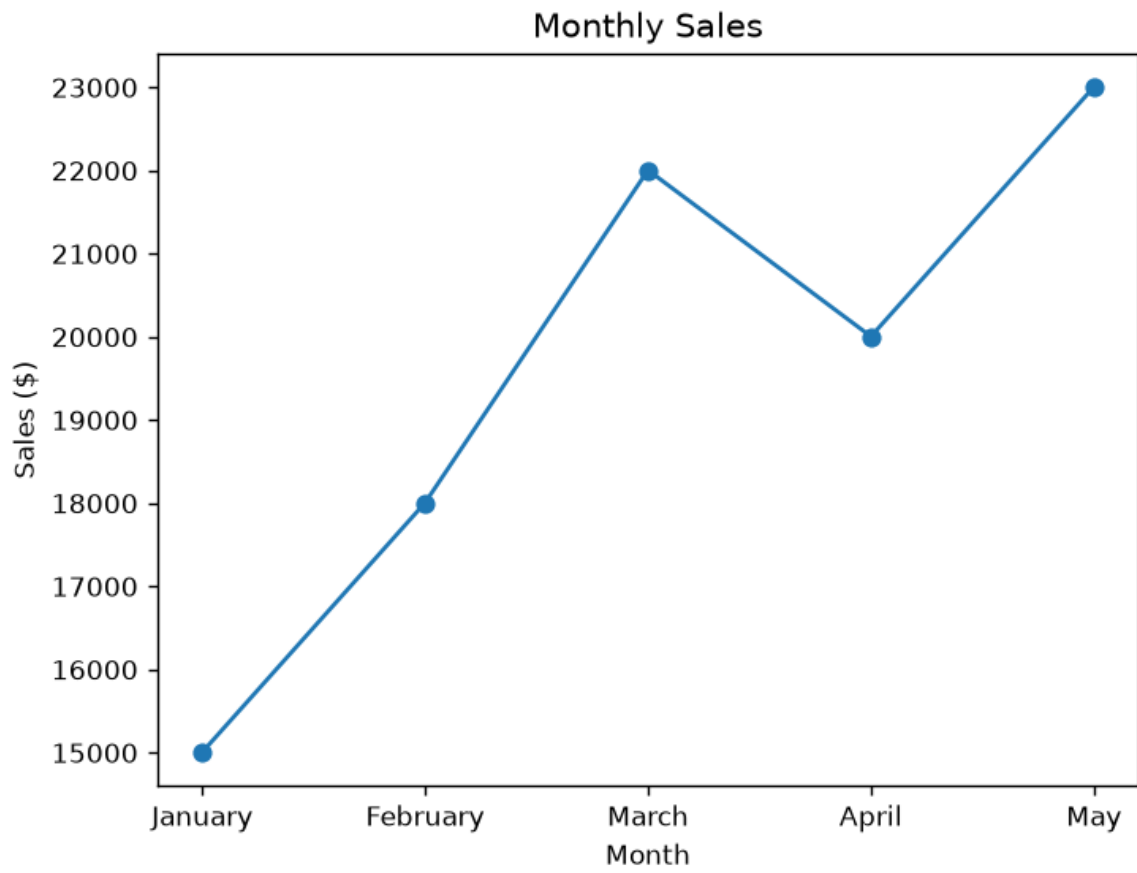
```
plt.xlabel("Advertising Budget")
```

```
plt.ylabel("Monthly Sales")
```

```
plt.show()
```

```
print("Insight: As the advertising budget increases, monthly sales also tend to increase.")
```

Output:



Set 2:Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.
3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.
4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

Python code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("Book2.csv")
```

```
# 1. Histogram
```

```
plt.hist(data["Age"], bins=5)
```

```
plt.title("Distribution of Customer Ages")
```

```
plt.xlabel("Age")
```

```
plt.ylabel("Frequency")
```

```
plt.show()
```

2. Pie Chart

```
score = data["Satisfaction Score"].value_counts()
```

```
plt.pie(score, labels=score.index, autopct="%1.1f%%")
```

```
plt.title("Customer Satisfaction Scores")
```

```
plt.show()
```

3. Stacked Bar Chart

```
data["Age Group"] = pd.cut(data["Age"],
```

```
    bins=[20, 30, 40, 50],
```

```
    labels=["20-30", "31-40", "41-50"])
```

```
table = pd.crosstab(data["Age Group"], data["Satisfaction Score"])
```

```
table.plot(kind="bar", stacked=True)
```

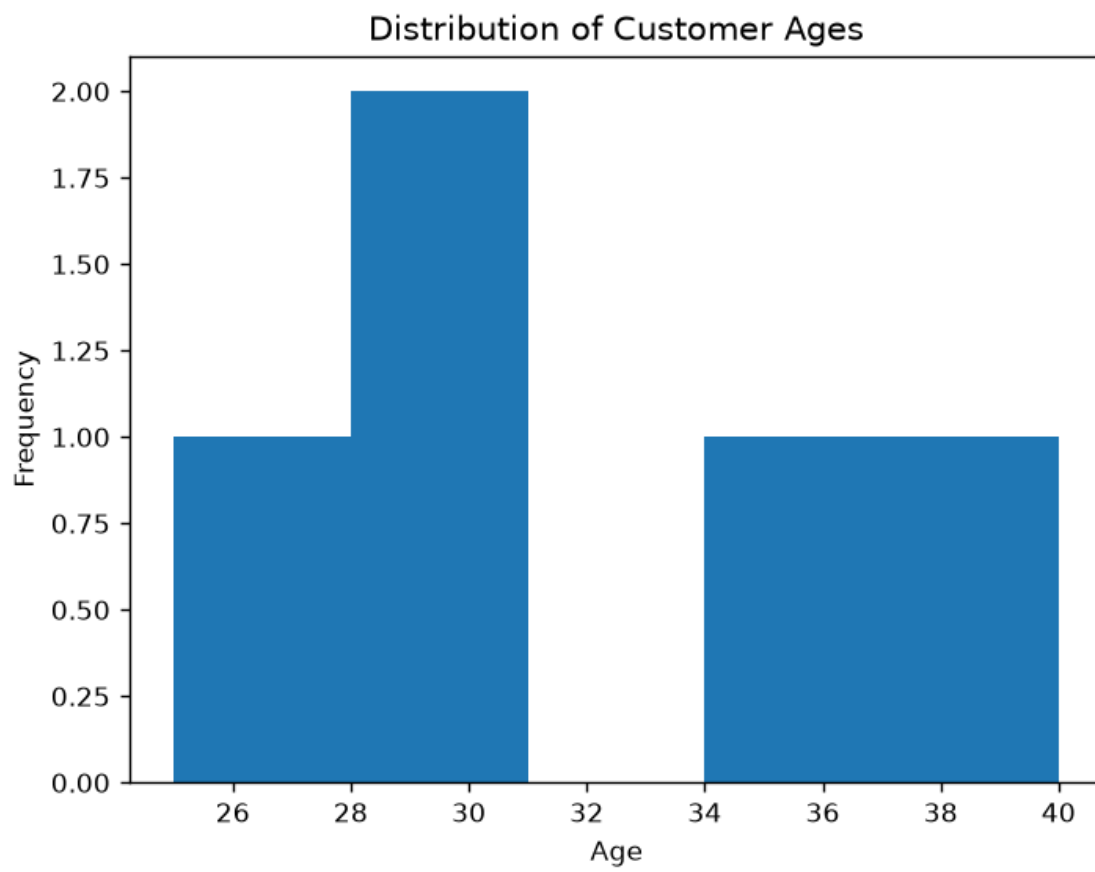
```
plt.title("Satisfaction Score by Age Group")
```

```
plt.xlabel("Age Group")
```

```
plt.ylabel("Count")
```

```
plt.show()
```

Output:



Set 3: Employee Performance Evaluation

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.
4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration

Python program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("Book3.csv")
```

```
# 1. Line Chart
```

```
plt.plot(data["Employee ID"], data["Performance Score"], marker="o", label="Performance")
```

```
plt.title("Employee Performance Trend")
```

```
plt.xlabel("Employee ID")
```

```
plt.ylabel("Performance Score")
```

```
plt.legend()
```

```
plt.show()
```

2. Bar Chart

```
dept = data["Department"].value_counts()
```

```
plt.bar(dept.index, dept.values)
```

```
plt.title("Employees by Department")
```

```
plt.xlabel("Department")
```

```
plt.ylabel("Number of Employees")
```

```
plt.show()
```

3. Scatter Plot

```
plt.scatter(data["Years of Service"], data["Performance Score"])
```

```
plt.title("Years of Service vs Performance Score")
```

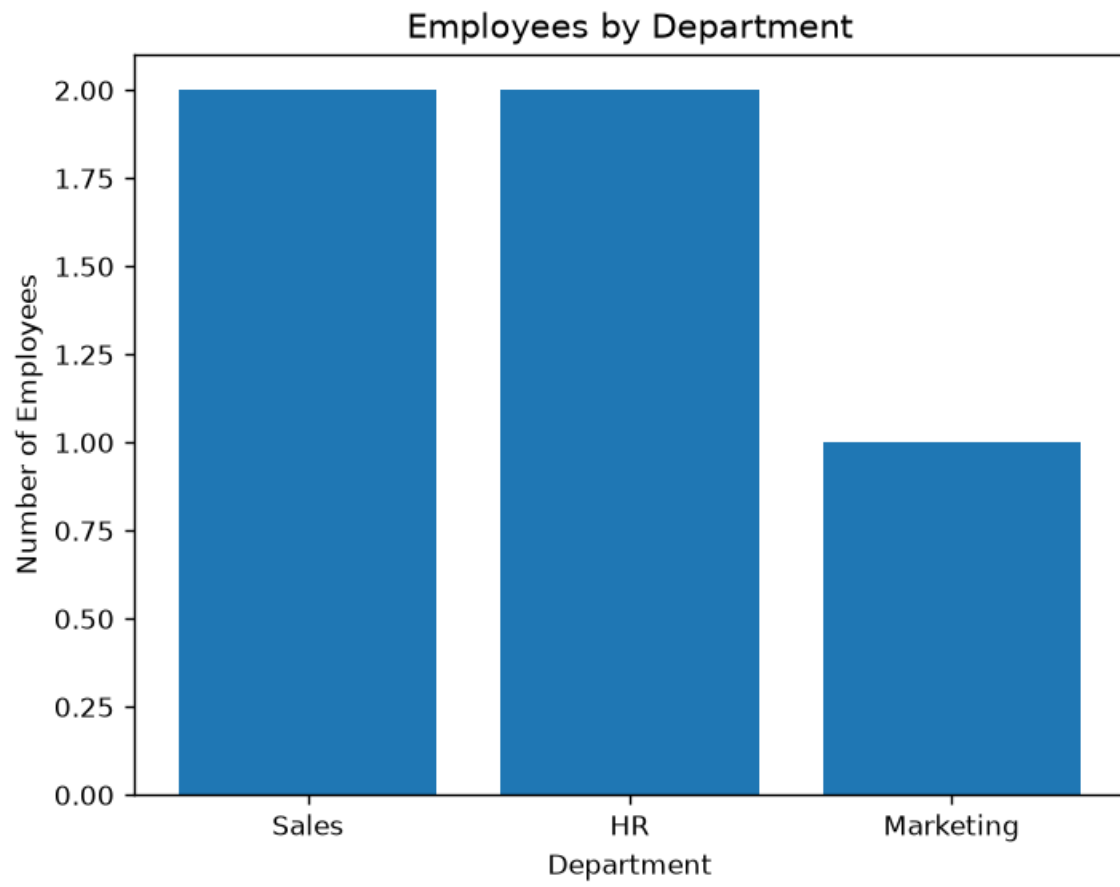
```
plt.xlabel("Years of Service")
```

```
plt.ylabel("Performance Score")
```

```
plt.show()
```

```
print("Insight: Employees with more years of service generally tend to have higher  
performance scores, although there are some variations.")
```


Output:



Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.
4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Python program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("product_inventory.csv")
```

```
# 1. Bar Chart
```

```
plt.bar(data["Product Name"], data["Quantity Available"])
```

```
plt.title("Product Inventory")
```

```
plt.xlabel("Product Name")
```

```
plt.ylabel("Quantity Available")
```

```
plt.show()
```

```
# 2. Stacked Bar Chart
```

```
table = pd.crosstab(data["Category"], data["Product Name"],
```

```
                    values=data["Quantity Available"],
```

```
                    aggfunc="sum").fillna(0)
```

```
table.plot(kind="bar", stacked=True)
```

```
plt.title("Product Quantity by Category")
```

```
plt.xlabel("Category")
```

```
plt.ylabel("Quantity")
```

```
plt.show()
```

```
# 3. Scatter Plot
```

```
plt.scatter(data["Price"], data["Quantity Available"])
```

```
plt.title("Price vs Quantity Available")
```

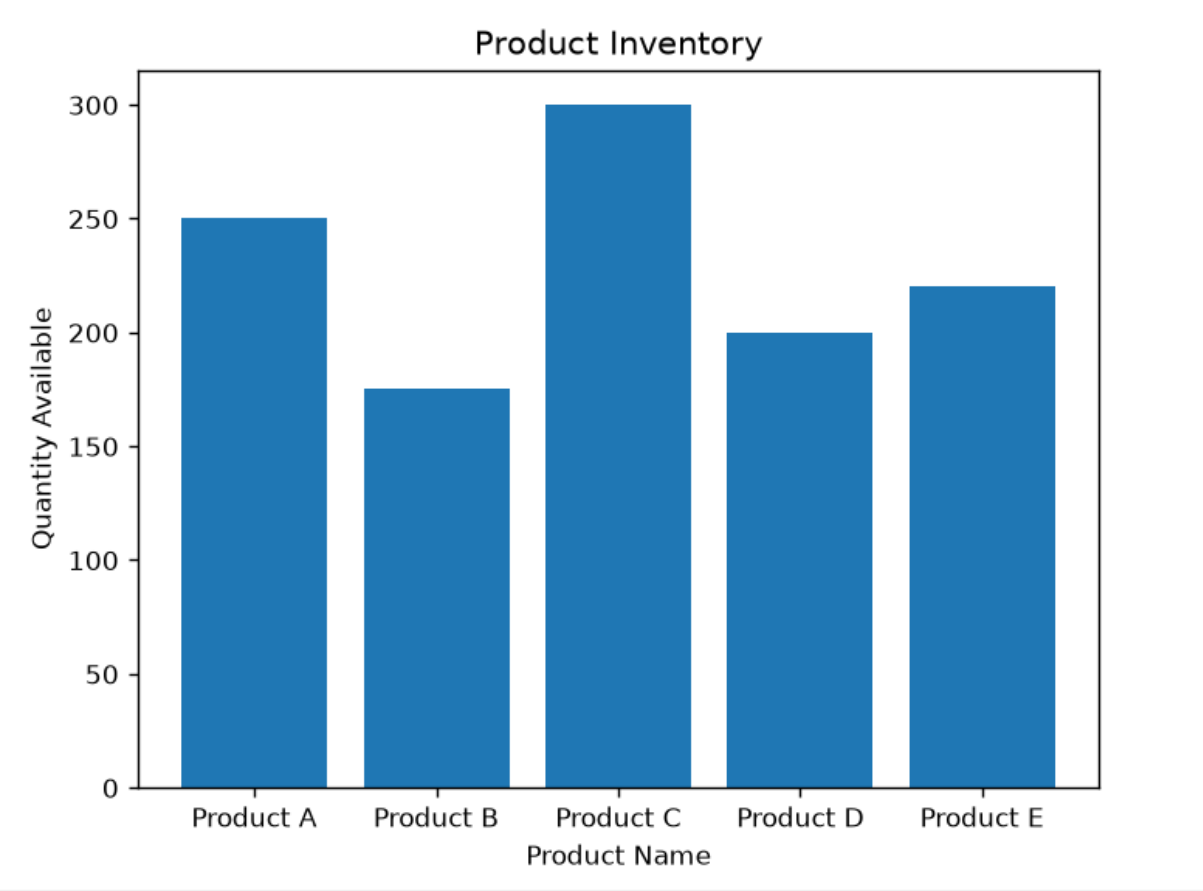
```
plt.xlabel("Price")
```

```
plt.ylabel("Quantity Available")
```

```
plt.show()
```

```
print("Insight: The scatter plot helps identify whether higher-priced products have higher or lower inventory levels.")
```

Output:



Set 5: Website Analytics

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.
4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Python program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read CSV file
data = pd.read_csv("Book5.csv")

# Convert Click-through Rate from percentage to number
data["Click-through Rate"] = data["Click-through Rate"].str.replace("%", "").astype(float)

# 1. Line Chart
plt.plot(data["Date"], data["Page Views"], marker="o")
plt.title("Daily Page Views")
plt.xlabel("Date")
plt.ylabel("Page Views")
plt.xticks(rotation=45)
```

```
plt.show()
```

```
# 2. Bar Chart (Top 5 Click-through Rates)
```

```
top = data.sort_values(by="Click-through Rate", ascending=False).head(5)
```

```
plt.bar(top["Date"], top["Click-through Rate"])
```

```
plt.title("Top Click-through Rates")
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Click-through Rate (%)")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

```
# 3. Stacked Area Chart
```

```
plt.stackplot(
```

```
    data["Date"],
```

```
    data["Likes"],
```

```
    data["Shares"],
```

```
    data["Comments"],
```

```
    labels=["Likes", "Shares", "Comments"]
```

```
)
```

```
plt.title("User Interactions")
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Count")
```

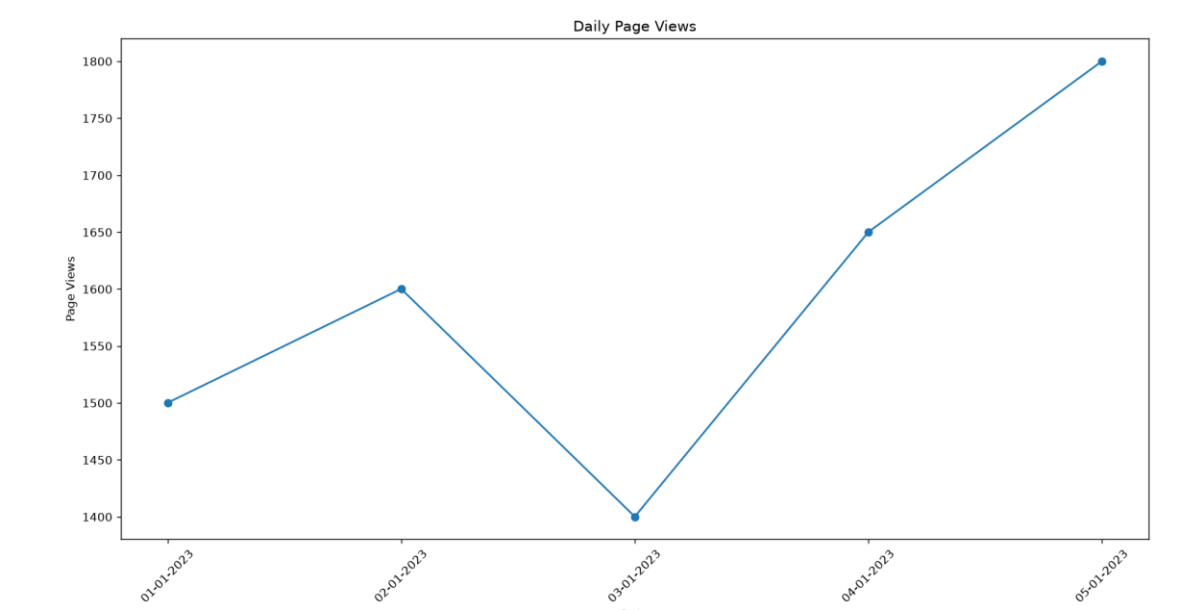
```
plt.legend()
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

```
print("Insight: The line chart shows page view trends, while the bar chart highlights the days  
with the highest click-through rates.")
```

Output:



Set 6

Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.
3. Build a table to show the monthly sales data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Python program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("monthly_product_sales.csv")
```

```
# 1. Grouped Bar Chart
```

```
x = range(len(data))
```

```
plt.bar([i-0.2 for i in x], data["January Sales"], width=0.2, label="January")
```



```
plt.bar(x, data["February Sales"], width=0.2, label="February")

plt.bar([i+0.2 for i in x], data["March Sales"], width=0.2, label="March")


plt.xticks(x, data["Product Name"])

plt.title("Monthly Product Sales")

plt.xlabel("Product")

plt.ylabel("Sales")

plt.legend()

plt.show()
```

2. Stacked Area Chart

```
plt.stackplot(

    ["January", "February", "March"],

    data["January Sales"],

    data["February Sales"],

    data["March Sales"],

    labels=data["Product Name"]

)
```

```
plt.title("Overall Sales Trend")

plt.xlabel("Month")

plt.ylabel("Sales")

plt.legend()
```

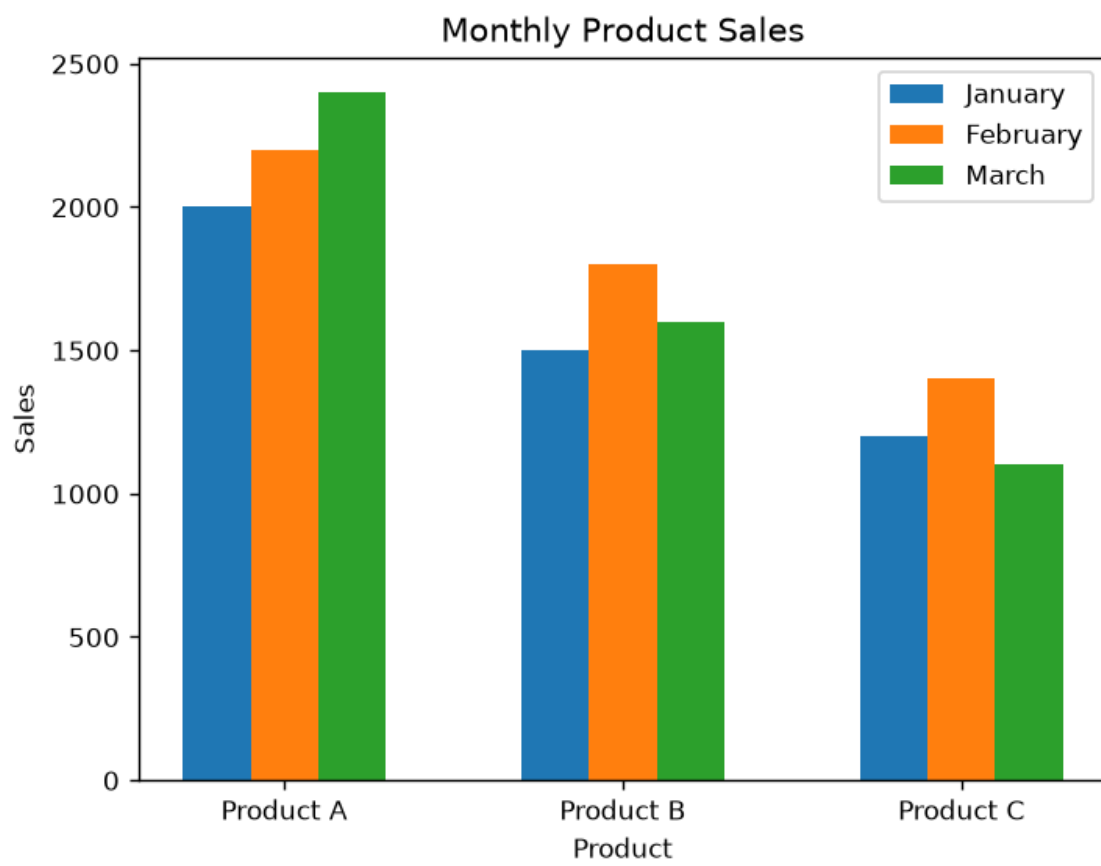
```
plt.show()
```

```
# 3. Table
```

```
print("Monthly Product Sales")
```

```
print(data)
```

Output:



Set 7 Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.
3. Build a table to show the demographic information for each customer. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Python Program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("Book7.csv")
```

```
# 1. Histogram of Age
```

```
plt.hist(data["Age"], bins=5)
```

```
plt.title("Age Distribution")
```

```
plt.xlabel("Age")
```

```
plt.ylabel("Frequency")
```

```
plt.show()
```

```
# 2. Bar Chart of Gender
```

```
gender = data["Gender"].value_counts()
```

```
plt.bar(gender.index, gender.values)
```

```
plt.title("Gender Distribution")
```

```
plt.xlabel("Gender")  
plt.ylabel("Count")  
plt.show()
```

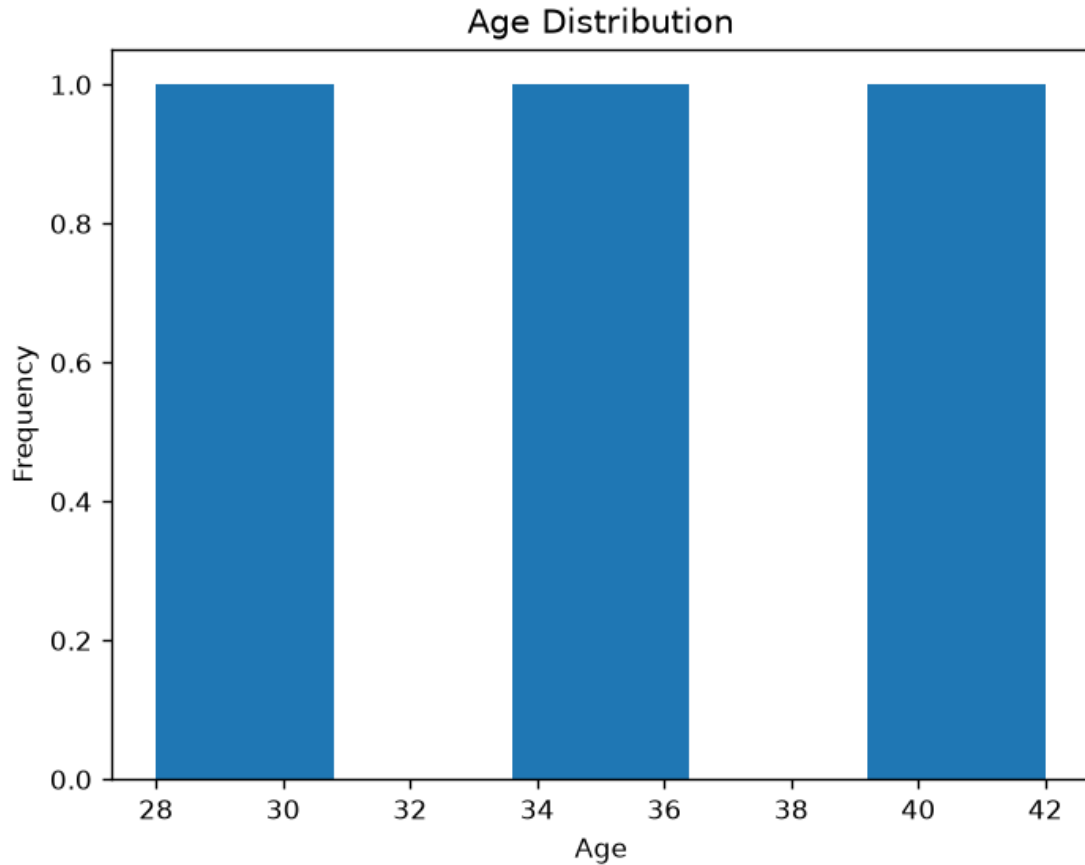
3. Scatter Plot (Age vs Income)

```
plt.scatter(data["Age"], data["Income (in $)"])  
plt.title("Age vs Income")  
plt.xlabel("Age")  
plt.ylabel("Income ($)")  
plt.show()
```

4. Table

```
print("Customer Data")  
print(data)
```

Output:



Set 8 Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a table to display the performance data for each employee. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Python program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("Book8.csv")
```

```
# 1. Line Chart
```

```
plt.plot(data["Employee ID"], data["Performance Score"], marker="o")
```

```
plt.title("Employee Performance Trend")
```

```
plt.xlabel("Employee ID")
```

```
plt.ylabel("Performance Score")
```

```
plt.grid(True)
```

```
plt.show()
```

2. Bar Chart

```
dept = data["Department"].value_counts()
```

```
plt.bar(dept.index, dept.values)
```

```
plt.title("Employees by Department")
```

```
plt.xlabel("Department")
```

```
plt.ylabel("Number of Employees")
```

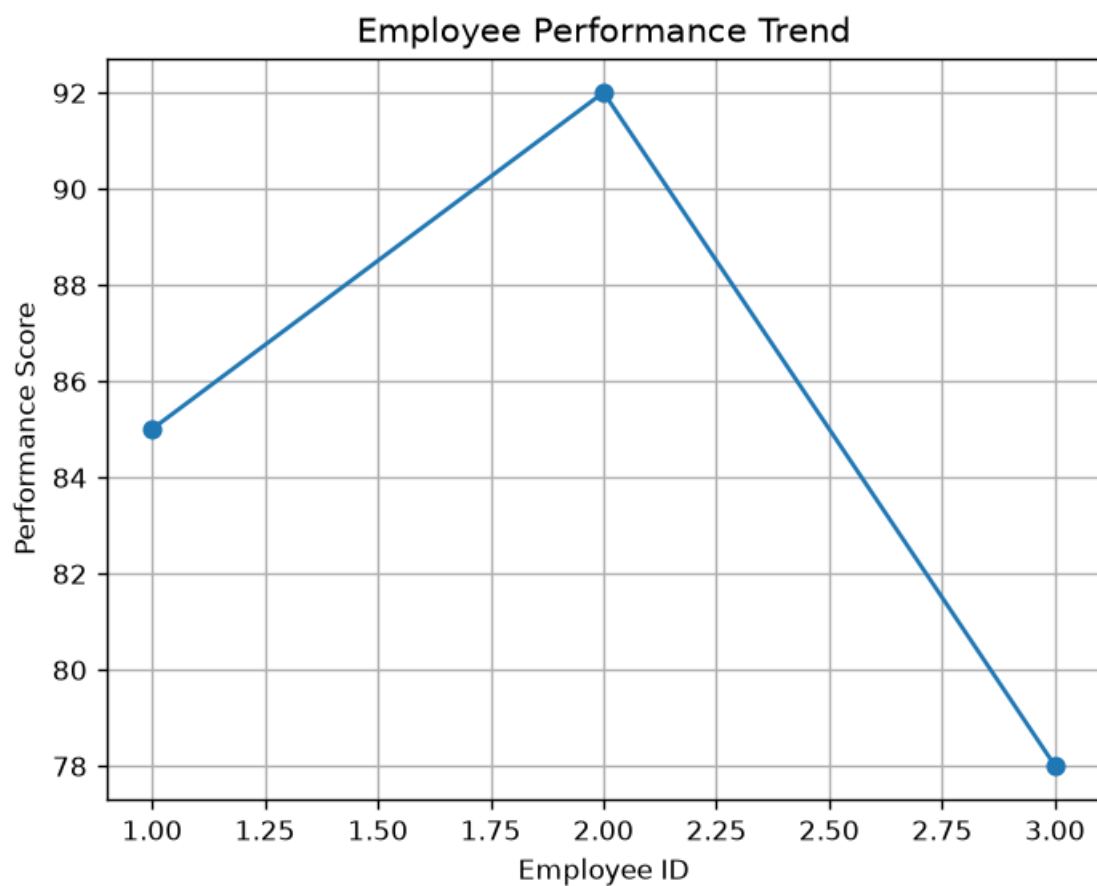
```
plt.show()
```

3. Display Table

```
print("Employee Performance Data")
```

```
print(data)
```

Output:



set 9

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

1. Import the dataset in **R**. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.
2. Create a scatter plot **using R programming** between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.
3. Using **R programming**, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.
4. Import Online_Learning_Activity.csv **into Tableau**. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

Python program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("Book9.csv")
```

1. Histogram of Quiz Score

```
plt.hist(data["Quiz_Score"], bins=5)

plt.title("Quiz Score Distribution")

plt.xlabel("Quiz Score")

plt.ylabel("Frequency")

plt.show()
```

2. Boxplot of Quiz Score by Course

```
data.boxplot(column="Quiz_Score", by="Course")

plt.title("Quiz Score by Course")

plt.suptitle("")

plt.xlabel("Course")

plt.ylabel("Quiz Score")

plt.show()
```

3. Scatter Plot (Study Time vs Quiz Score)

```
plt.scatter(

    data["Study_Time (hrs)"],

    data["Quiz_Score"],

    s=data["Videos_Watched"]*20,

    alpha=0.5

)

plt.title("Study Time vs Quiz Score")

plt.xlabel("Study Time (hrs)")
```



```
plt.ylabel("Quiz Score")
```

```
plt.show()
```

```
# 4. Monthly Average Quiz Score
```

```
data["Login_Date"] = pd.to_datetime(data["Login_Date"])
```

```
data["Month"] = data["Login_Date"].dt.to_period("M")
```

```
avg = data.groupby("Month")["Quiz_Score"].mean()
```

```
# Moving Average
```

```
ma = avg.rolling(2).mean()
```

```
plt.plot(avg.index.astype(str), avg.values, marker="o", label="Average Score")
```

```
plt.plot(avg.index.astype(str), ma.values, marker="o", label="Moving Average")
```

```
plt.title("Average Monthly Quiz Score")
```

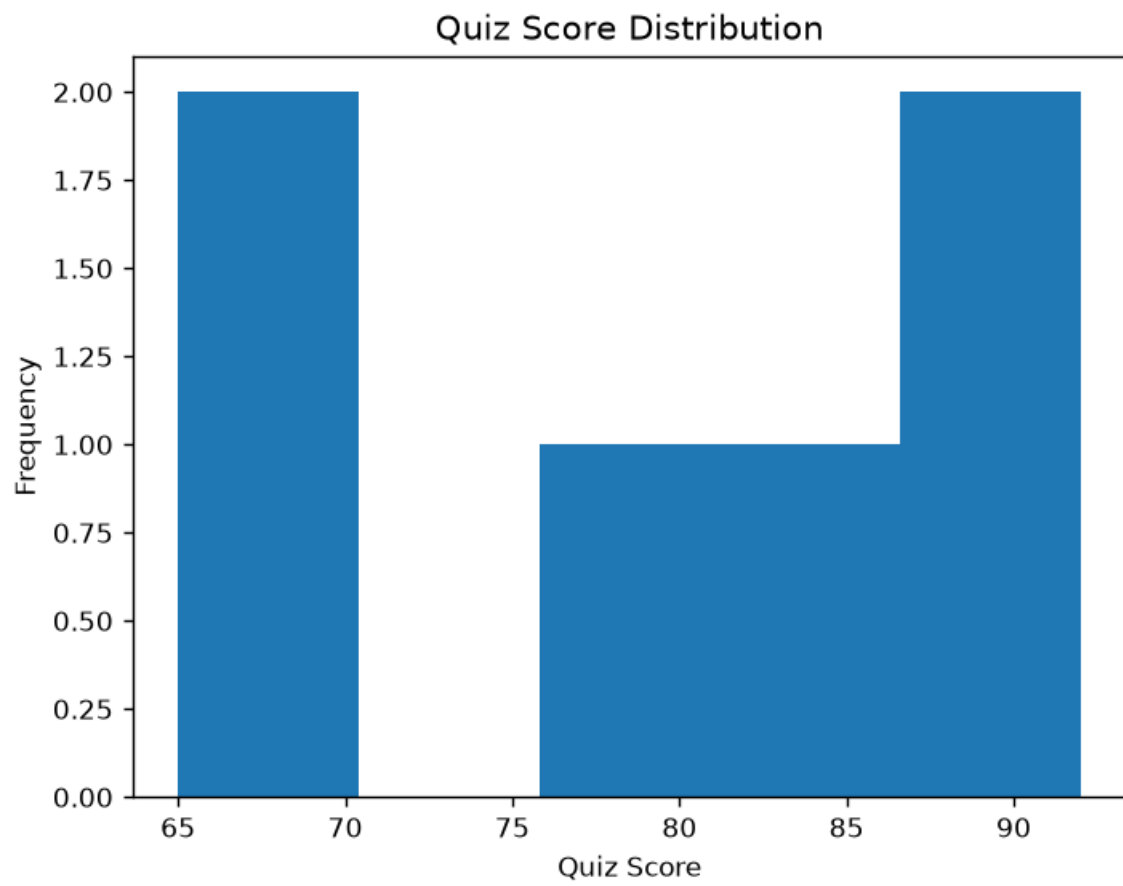
```
plt.xlabel("Month")
```

```
plt.ylabel("Average Quiz Score")
```

```
plt.legend()
```

```
plt.show()
```

Output:



Set 10 Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.
2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.
3. Build a table to show the survey response data for each survey. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Python program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("Book10.csv")
```

```
# 1. Grouped Bar Chart (Question 1 Responses)
```

```
q1 = data["Question 1"].value_counts()
```

```
plt.bar(q1.index, q1.values)
```

```
plt.title("Distribution of Question 1 Responses")
```

```
plt.xlabel("Answer")
```

```
plt.ylabel("Count")
```

```
plt.show()
```

```
# 2. Stacked Bar Chart (All Questions)
```

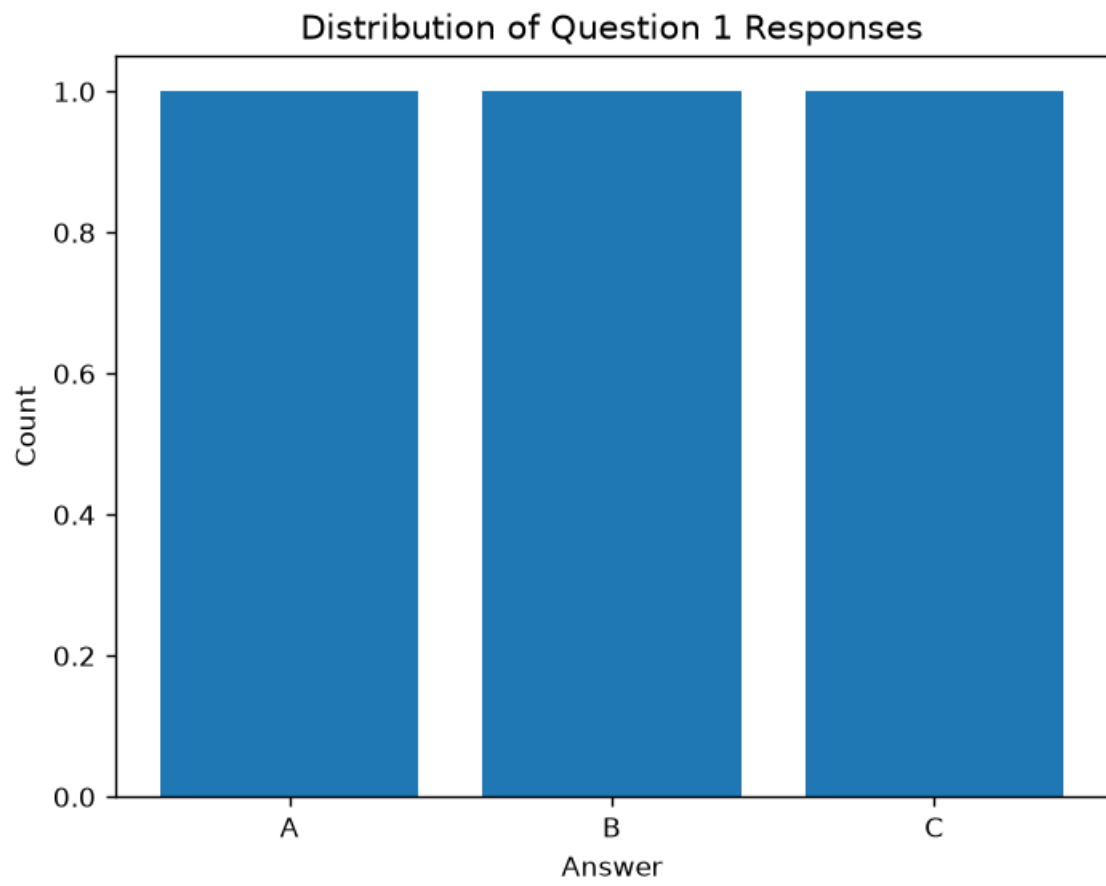
```
table = pd.DataFrame({  
    "Question 1": data["Question 1"].value_counts(),  
    "Question 2": data["Question 2"].value_counts(),  
    "Question 3": data["Question 3"].value_counts()  
}).fillna(0)
```

```
table.plot(kind="bar", stacked=True)  
plt.title("Survey Responses")  
plt.xlabel("Answer")  
plt.ylabel("Count")  
plt.show()
```

```
# 3. Display Table
```

```
print("Survey Response Data")  
print(data)
```

Output:



Set 11

Product Category Analysis

Dataset: Product Categories

Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.
2. Build a table to display the sales data for each product category. Label the table elements.
3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.
4. Create a pie chart to represent the distribution of sales across product categories. Include labels.

Python program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("Book11.csv")
```

```
# 1. Funnel Chart (Horizontal Bar Chart)
```

```
plt.barh(data["Category"], data["Sales (in $)"])
```

```
plt.title("Sales Conversion by Product Category")
```

```
plt.xlabel("Sales ($)")
```

```
plt.ylabel("Category")
```

```
plt.show()
```

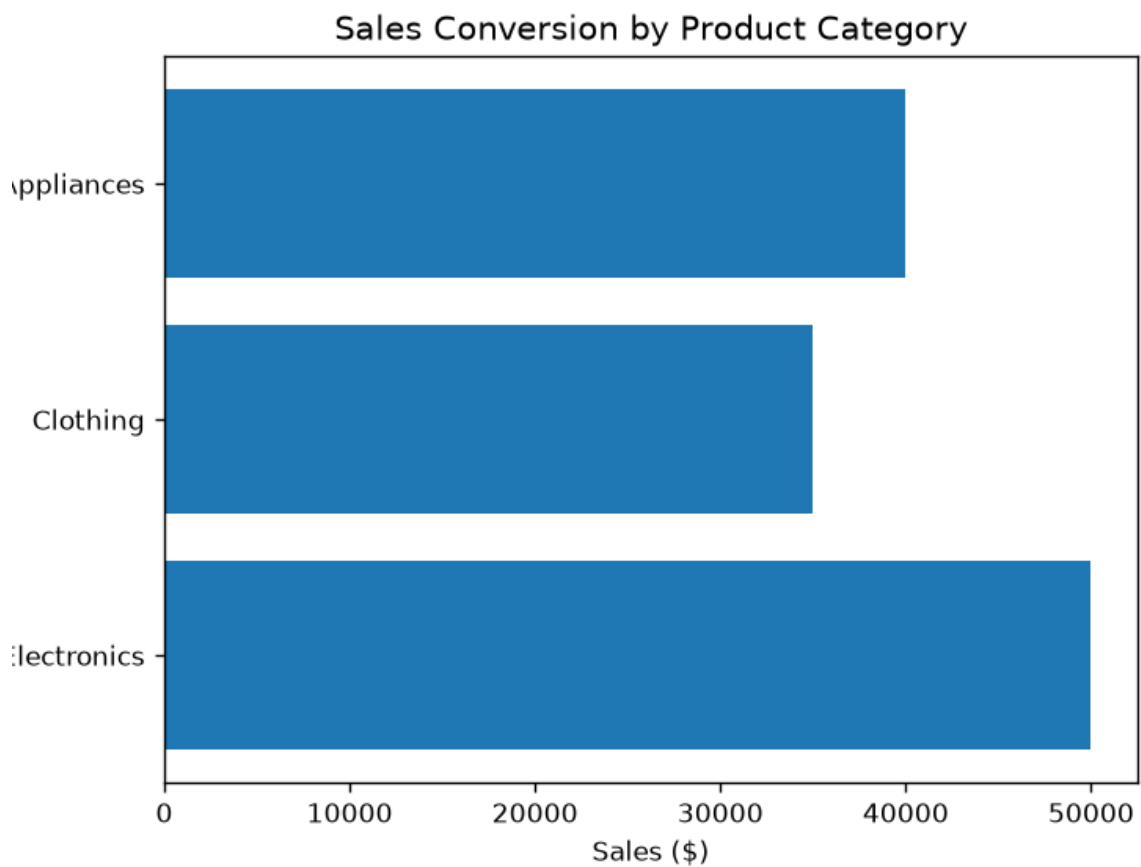
```
# 2. Display Table
```

```
print("Product Category Sales")
print(data)
```

3. Pie Chart

```
plt.pie(
    data["Sales (in $)"],
    labels=data["Category"],
    autopct="%1.1f%%"
)
plt.title("Sales Distribution by Product Category")
plt.show()
```

Output:



Set 12 :Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Build a table to show the website traffic data for each day. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

Python program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("Book12.csv")
```

```
# Convert Click-through Rate (%) to numeric
```

```
data["Click-through Rate"] = data["Click-through Rate"].str.replace("%", "").astype(float)
```

```
# 1. Line Chart
```

```
plt.plot(data["Date"], data["Page Views"], marker="o")
```

```
plt.title("Daily Page Views")
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Page Views")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```



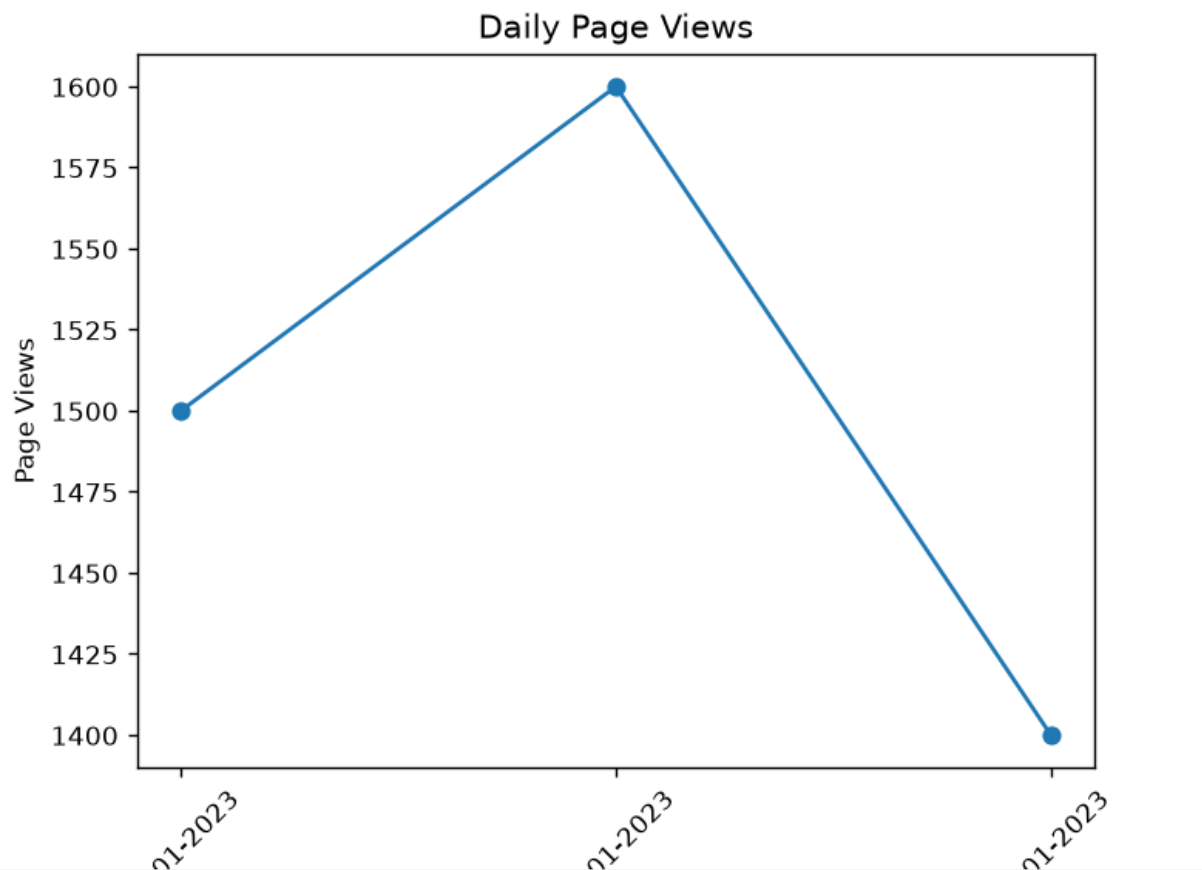
```
# 2. Bar Chart (Top N Days with Highest Click-through Rates)
top = data.sort_values(by="Click-through Rate", ascending=False)
```

```
plt.bar(top["Date"], top["Click-through Rate"])
plt.title("Top Click-through Rates")
plt.xlabel("Date")
plt.ylabel("Click-through Rate (%)")
plt.xticks(rotation=45)
plt.show()
```

```
# 3. Display Table
```

```
print("Website Traffic Data")
print(data)
```

Output:



13. Dataset: Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.
2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.
3. Build a table to display the geographic data for each city. Label the table elements.
4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Python program:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read CSV file
```

```
data = pd.read_csv("Book13.csv")
```

```
# 1. Simple City Plot (Map Placeholder)
```

```
plt.plot(data["City"], [1]*len(data), "o")
```

```
plt.title("City Distribution")
```

```
plt.xlabel("City")
```

```
plt.yticks([])
```

```
plt.show()
```

```
# 2. Scatter Plot
```

```
plt.scatter(data["Avg. Temperature"], data["Population"])
```

```
plt.title("Temperature vs Population")
```

```
plt.xlabel("Average Temperature")
```

```
plt.ylabel("Population")
```

```
plt.show()
```

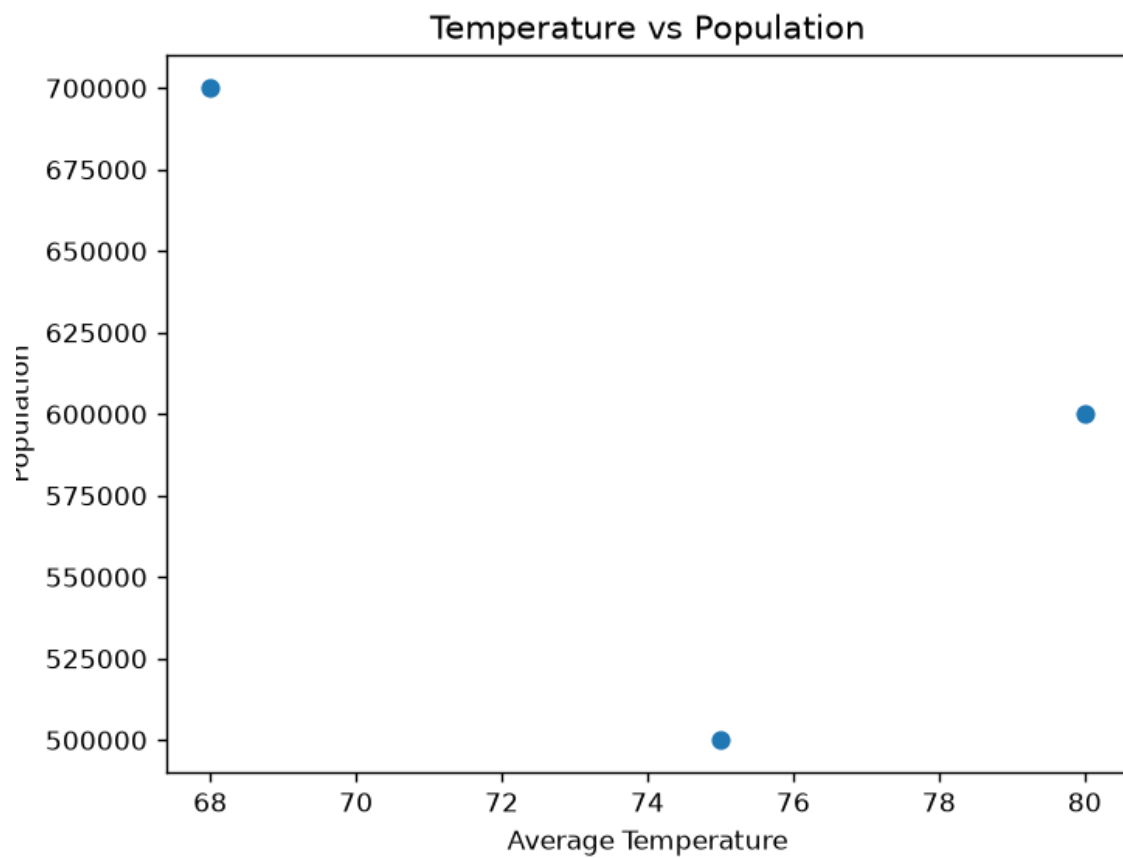
```
print("Insight: The scatter plot helps compare population size with average temperature.")
```

```
# 3. Display Table
```

```
print("Geographic Data")
```

```
print(data)
```

Output:



14. Dataset: Survey Results

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.
2. Generate a radar chart to visualize the overall pattern of responses across all three questions.
3. Build a table to show the survey response data for each respondent. Label the table elements.
4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

Python program:

```
import pandas as pd
import matplotlib.pyplot as plt
import math

# Read CSV file
data = pd.read_csv("Book14.csv")

# 1. Stacked Bar Chart
table = pd.DataFrame({
    "Question 1": data["Question 1"].value_counts(),
    "Question 2": data["Question 2"].value_counts(),
    "Question 3": data["Question 3"].value_counts()
}).fillna(0)

table.plot(kind="bar", stacked=True)
plt.title("Survey Response Distribution")
plt.xlabel("Answer")
```

```
plt.ylabel("Count")
plt.show()
```

```
# 2. Radar Chart
```

```
labels = ["Question 1", "Question 2", "Question 3"]
```

```
values = [
    data["Question 1"].value_counts().max(),
    data["Question 2"].value_counts().max(),
    data["Question 3"].value_counts().max()
]
```

```
# Create angles using math
```

```
angles = []
```

```
for i in range(len(labels)):
```

```
    angles.append(2 * math.pi * i / len(labels))
```

```
angles.append(angles[0])
```

```
values.append(values[0])
```

```
ax = plt.subplot(111, polar=True)
```

```
ax.plot(angles, values, marker="o")
```

```
ax.fill(angles, values, alpha=0.3)
```

```
ax.set_xticks(angles[:-1])
```

```
ax.set_xticklabels(labels)
```

```
plt.title("Survey Response Pattern")
```

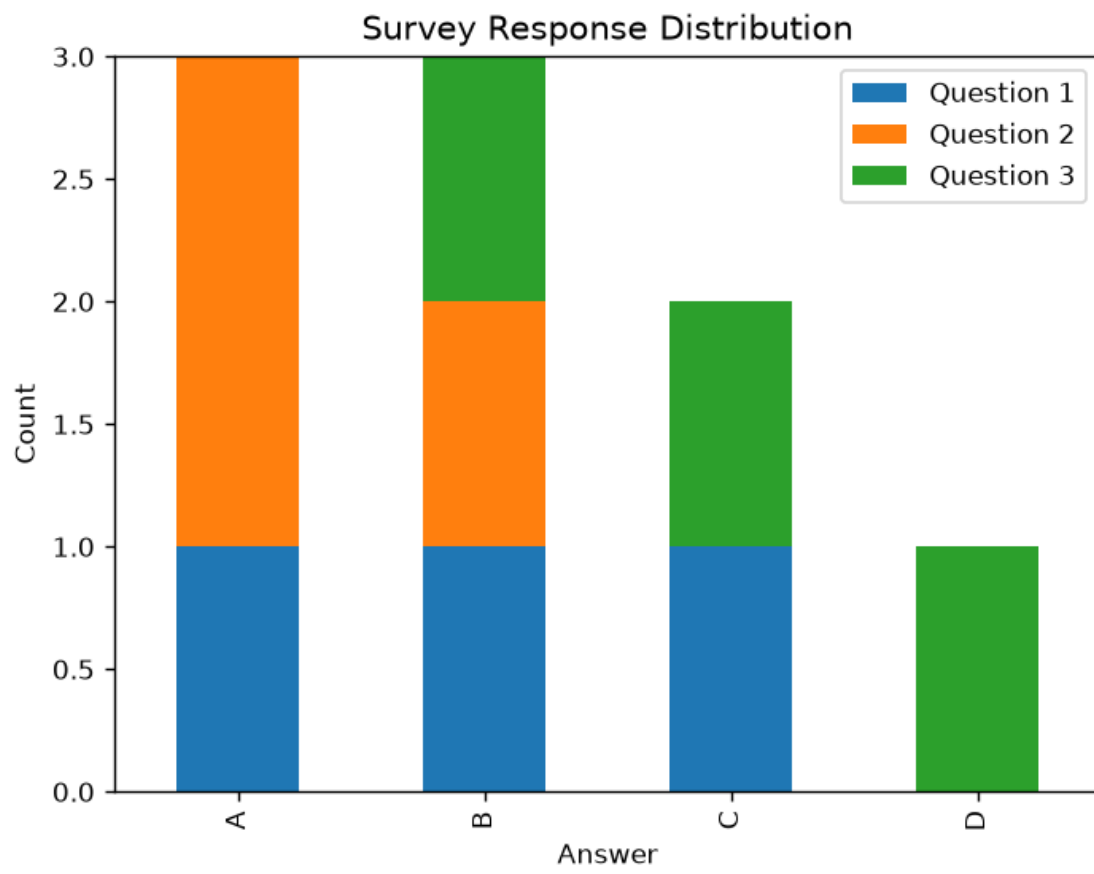
```
plt.show()
```

```
# 3. Display Table
```

```
print("Survey Results")
```

```
print(data)
```

Output:



Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.
3. Build a table to show the monthly sales data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Python program:

```
import csv
```

```
import matplotlib.pyplot as plt
```

```
products = []
```

```
jan = []
```

```
feb = []
```

```
mar = []
```

```
with open("Book15.csv", "r") as file:
```

```
    data = csv.reader(file)
```

```
    next(data)
```

```
    for row in data:
```

```
        products.append(row[1])    # Product Name
```

```
        jan.append(int(row[2]))    # January Sales
```

```
        feb.append(int(row[3]))    # February Sales
```

```
mar.append(int(row[4]))    # March Sales
```

```
# Grouped Bar Chart
```

```
x = [0, 1, 2]
```

```
w = 0.25
```

```
plt.bar([i-w for i in x], jan, width=w, label="January")
```

```
plt.bar(x, feb, width=w, label="February")
```

```
plt.bar([i+w for i in x], mar, width=w, label="March")
```

```
plt.xticks(x, products)
```

```
plt.xlabel("Products")
```

```
plt.ylabel("Sales")
```

```
plt.title("Monthly Product Sales")
```

```
plt.legend()
```

```
plt.show()
```

```
# Stacked Area Chart
```

```
months = ["January", "February", "March"]
```

```
plt.stackplot(months, jan, feb, mar, labels=products)
```

```
plt.xlabel("Months")
```

```
plt.ylabel("Sales")
```

```
plt.title("Overall Sales Trend")
```

```
plt.legend()
```

```
plt.show()
```

```
# Table
```

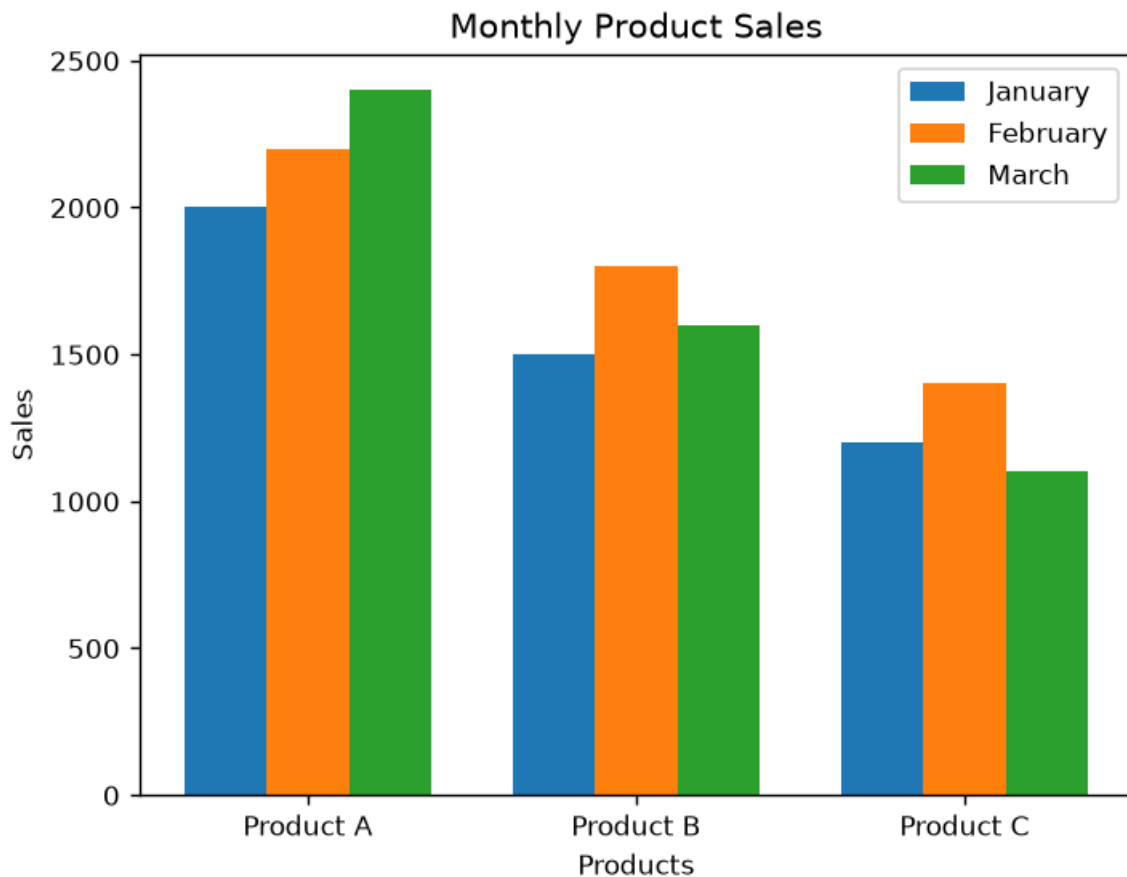
```
print("\nMonthly Product Sales")
```

```
print("Product\t\tJanuary\tFebruary\tMarch")
```

```
for i in range(len(products)):
```

```
    print(products[i], "\t", jan[i], "\t", feb[i], "\t", mar[i])
```


Output:



Set 16: Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.
3. Build a table to show the demographic information for each customer. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Python program:

```
import csv
```

```
import matplotlib.pyplot as plt
```

```
id, age, gender = [], [], []
```

```
with open("Book16.csv") as file:
```

```
    data = csv.reader(file)
```

```
    next(data)
```

```
    for row in data:
```

```
        id.append(row[0])
```

```
        age.append(int(row[1]))
```

```
        gender.append(row[2])
```

```
# 1. Bar Chart
```

```
plt.bar(id, age)
```

```
plt.title("Customer Ages")
```

```
plt.xlabel("Customer ID")
```

```
plt.ylabel("Age")
```

```
plt.show()
```

```
# 2. Pie Chart
```

```
male = gender.count("Male")
```

```
female = gender.count("Female")
```

```
plt.pie([male, female], labels=["Male", "Female"], autopct="%1.1f%%")
```

```
plt.title("Gender Distribution")
```

```
plt.show()
```

```
# 3. Table
```

```
print("ID\tAge\tGender")
```

```
for i in range(len(id)):
```

```
    print(id[i], "\t", age[i], "\t", gender[i])
```

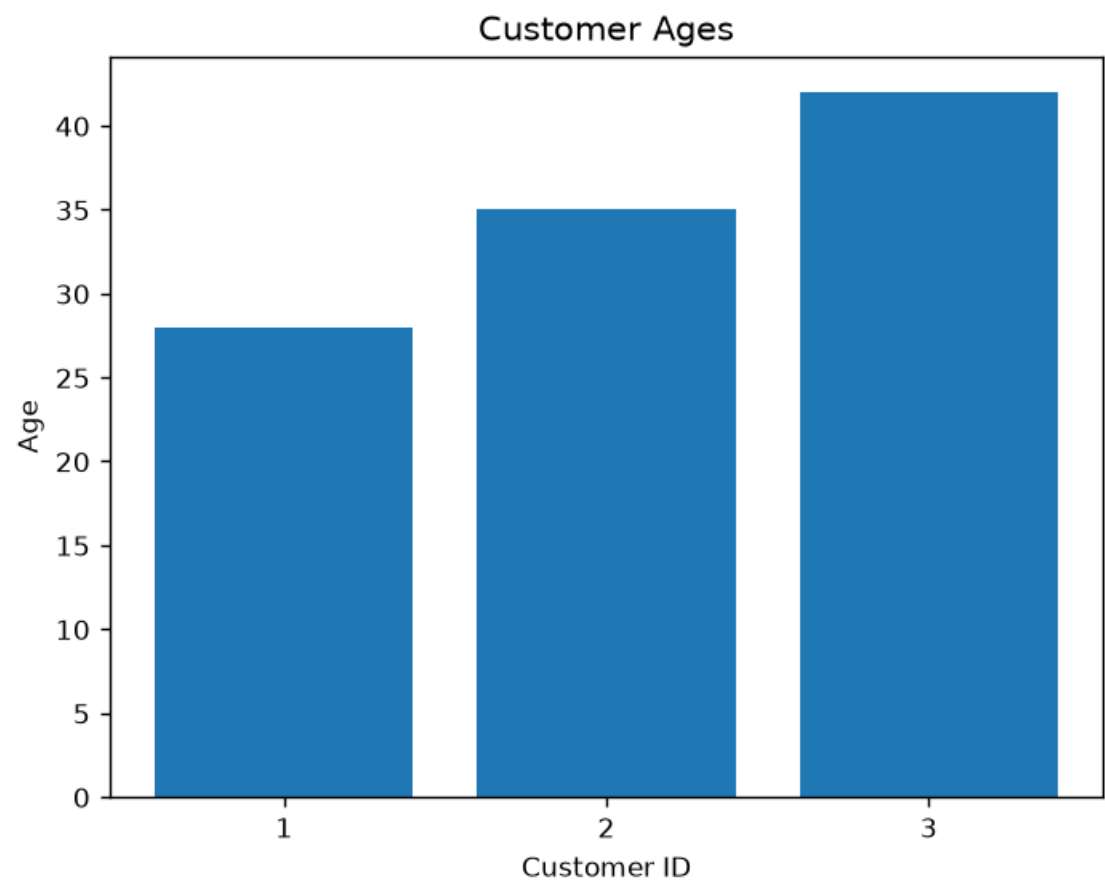
```
# 4. Tableau Dashboard
```

```
print("\nImport Customer_Demographics.csv into Tableau")
```

```
print("Create Bar Chart + Pie Chart + Table")
```

```
print("Combine them into a Dashboard")
```

Output:



Set 17: Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

- Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
- Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
- Build a table to display the performance data for each employee. Label the table elements.

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Python program:

```
import csv
```

```
import matplotlib.pyplot as plt
```

```
emp, dept, year, score = [], [], [], []
```

```
with open("Book17.csv") as file:
```

```
    data = csv.reader(file)
```

```
    next(data)
```

```
    for row in data:
```

```
        emp.append(row[0])
```

```
        dept.append(row[1])
```

```
        year.append(int(row[2]))
```

```
        score.append(int(row[3]))
```

```
# 1. Line Chart
```

```
plt.plot(year, score, marker='o')
```

```
plt.title("Employee Performance")
```

```
plt.xlabel("Years of Service")
```

```
plt.ylabel("Performance Score")
```

```
plt.show()
```

```
# 2. Bar Chart
```

```
count = {}
```

```
for d in dept:
```

```
    count[d] = count.get(d, 0) + 1
```

```
plt.bar(count.keys(), count.values())
```

```
plt.title("Employees by Department")
```

```
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.show()
```

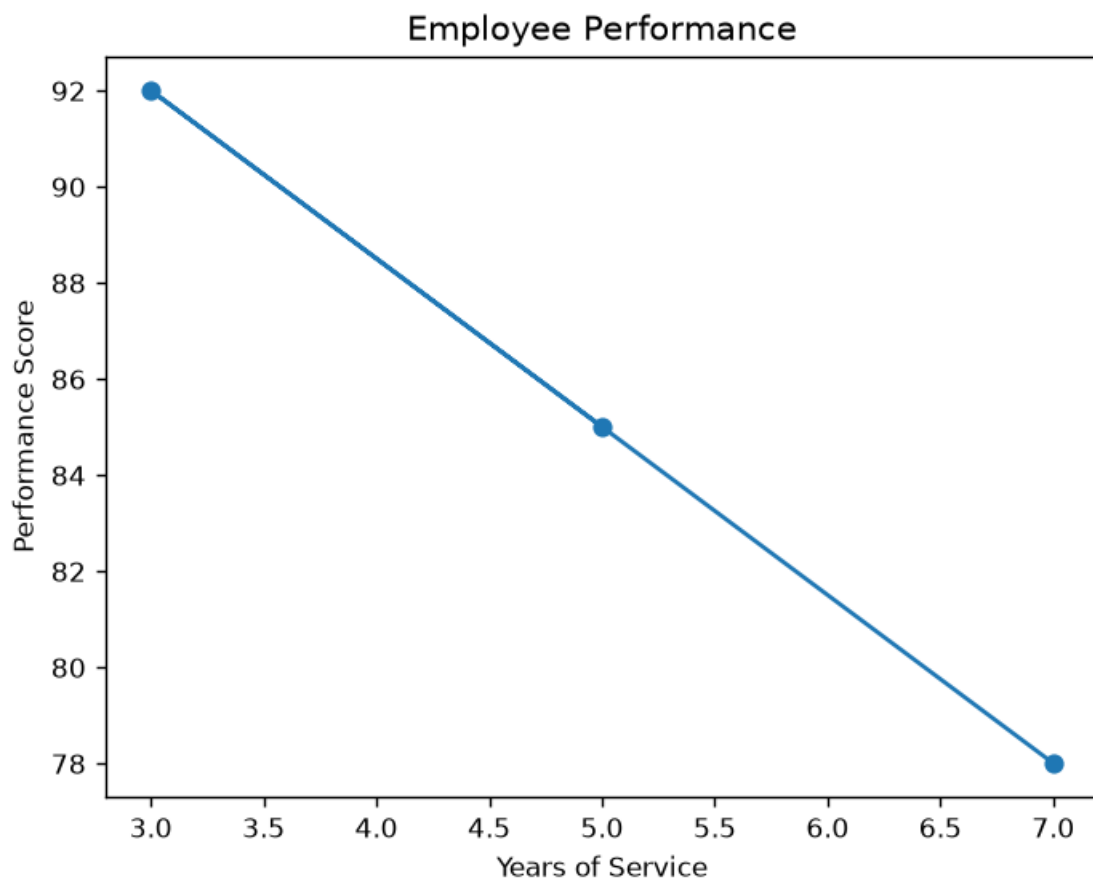
3. Table

```
print("ID\tDepartment\tYears\tScore")
for i in range(len(emp)):
    print(emp[i], "\t", dept[i], "\t\t", year[i], "\t", score[i])
```

4. Tableau Dashboard

```
print("\nImport Employee_Performance.csv into Tableau.")
print("Create Line Chart, Bar Chart and Table.")
print("Combine them into one Dashboard.")
```

Output:



Set 18: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in \$)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a table to show the inventory data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

Python program:

```
import csv
```

```
import matplotlib.pyplot as plt
```

```
pid = []
```

```
pname = []
```

```
qty = []
```

```
price = []
```

```
with open("Book19.csv", "r") as file:
```

```
    data = csv.reader(file)
```

```
    next(data)
```

```
    for row in data:
```

```
        if len(row) >= 4:
```

```
            pid.append(row[0])
```

```
            pname.append(row[1])
```

```
            qty.append(int(row[2]))
```

```
            price.append(int(row[3]))
```

1. Bar Chart

```
plt.bar(pname, qty)
plt.title("Product Inventory")
plt.xlabel("Product Name")
plt.ylabel("Quantity Available")
plt.show()
```

2. Stacked Bar Chart

```
plt.bar(pname, qty, label="Quantity")
plt.bar(pname, price, bottom=qty, label="Price")
plt.title("Inventory Details")
plt.xlabel("Product Name")
plt.ylabel("Value")
plt.legend()
plt.show()
```

3. Table

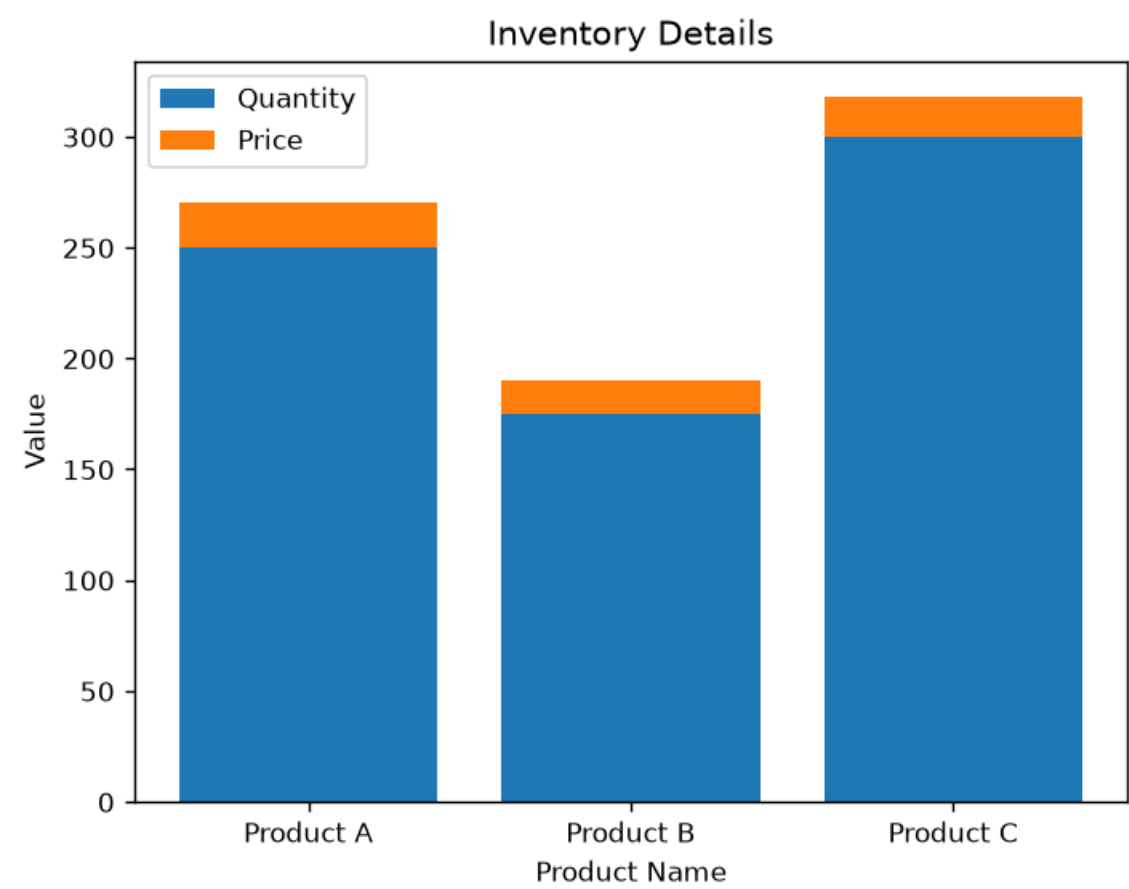
```
print("\nInventory Data")
print("ID\tProduct\t\tQuantity\tPrice")
```

```
for i in range(len(pid)):
    print(pid[i], "\t", pname[i], "\t", qty[i], "\t\t$", price[i])
```

4. Tableau Dashboard

```
print("\nImport Product_Inventory.csv into Tableau.")
print("Create:")
print("1. Bar Chart")
print("2. Stacked Bar Chart")
print("3. Table")
print("4. Combine them into a Dashboard")
```


Output:



Set 19: Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.
2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.
3. Build a table to show the survey response data for each survey. Label the table elements.

4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Python program:

```
import csv
```

```
import matplotlib.pyplot as plt
```

```
sid = []
```

```
q1 = []
```

```
q2 = []
```

```
q3 = []
```

```
with open("Book19(1).csv","r") as file:
```

```
    data = csv.reader(file)
```

```
    next(data)
```

```
    for row in data:
```

```
        sid.append(row[0])
```

```
        q1.append(row[1])
```

```
        q2.append(row[2])
```

```
        q3.append(row[3])
```

```
# 1. Grouped Bar Chart (Question 1)
```

```
a = q1.count("A")
```

```
b = q1.count("B")
```

```
c = q1.count("C")
```

```
plt.bar(["A","B","C"], [a,b,c])
```

```
plt.title("Question 1 Responses")
```

```
plt.xlabel("Answers")
```

```
plt.ylabel("Count")
```

```
plt.show()
```

```
# 2. Stacked Bar Chart
```

```
A = [q1.count("A"), q2.count("A"), q3.count("A")]
B = [q1.count("B"), q2.count("B"), q3.count("B")]
C = [q1.count("C"), q2.count("C"), q3.count("C")]
D = [q1.count("D"), q2.count("D"), q3.count("D")]
```

```
questions = ["Q1", "Q2", "Q3"]
```

```
plt.bar(questions, A, label="A")
plt.bar(questions, B, bottom=A, label="B")
```

```
bottom2 = [A[i]+B[i] for i in range(3)]
plt.bar(questions, C, bottom=bottom2, label="C")
```

```
bottom3 = [bottom2[i]+C[i] for i in range(3)]
plt.bar(questions, D, bottom=bottom3, label="D")
```

```
plt.title("Survey Responses")
plt.xlabel("Questions")
plt.ylabel("Count")
plt.legend()
plt.show()
```

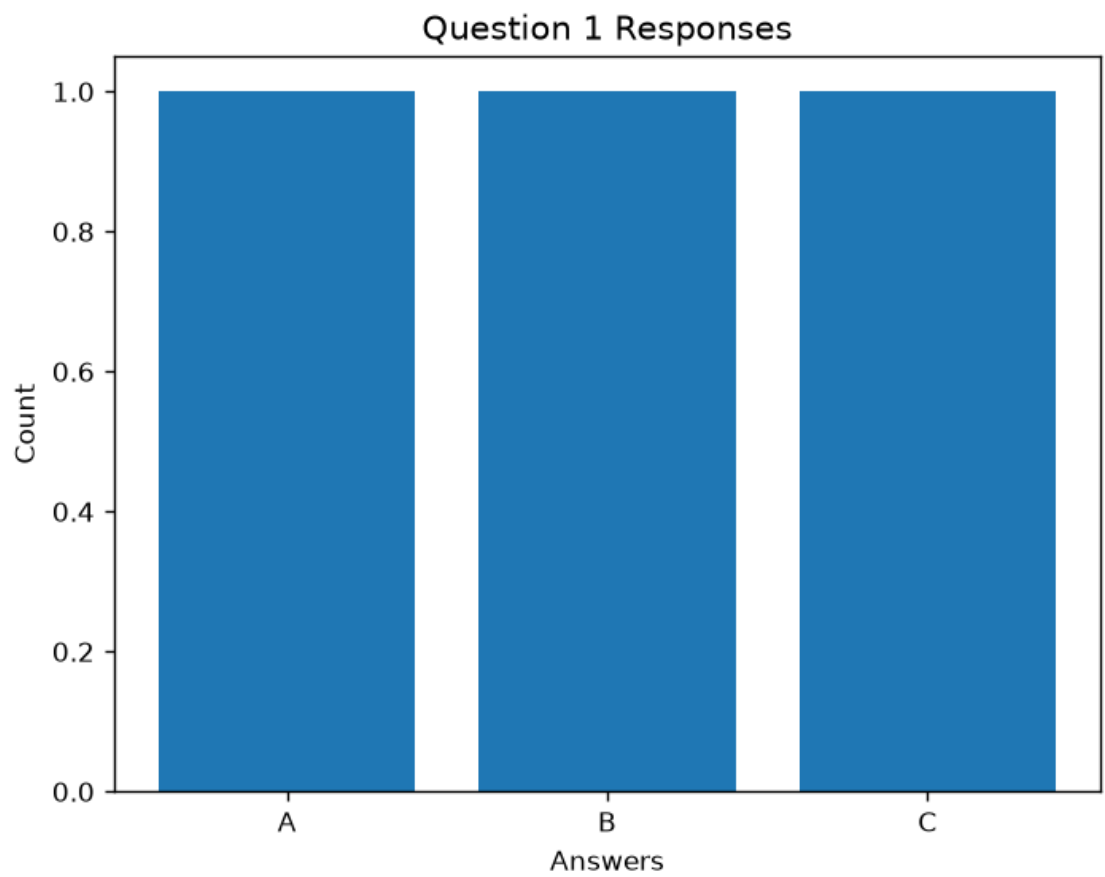
```
# 3. Table
```

```
print("Survey ID\tQ1\tQ2\tQ3")
for i in range(len(sid)):
    print(sid[i], "\t\t", q1[i], "\t", q2[i], "\t", q3[i])
```

```
# 4. Tableau Dashboard
```

```
print("\nImport Survey_Responses.csv into Tableau")
print("Create Grouped Bar Chart, Stacked Bar Chart and Table")
print("Combine them into one Dashboard")
```

Output:



Set 20: Stock Analysis

Dataset: Stock Prices

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.
2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.
3. Build a table to display the stock price data for each company over the given period. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

Python program:

```
import csv
```

```
import matplotlib.pyplot as plt
```

```
date = []
```

```
a = []
```

```
b = []
```

```
c = []
```

```
with open("Book20.csv","r") as file:
```

```
    data = csv.reader(file)
```

```
    next(data)
```

```
    for row in data:
```

```
        date.append(row[0])
```

```
        a.append(int(row[1]))
```

```
        b.append(int(row[2]))
```

```
        c.append(int(row[3]))
```

```
# 1. Line Chart
```

```
plt.plot(date, a, marker='o', label="Stock A")
```

```
plt.plot(date, b, marker='o', label="Stock B")
```

```
plt.plot(date, c, marker='o', label="Stock C")
```

```
plt.title("Stock Prices")
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Price")
```

```
plt.legend()
```

```
plt.show()
```

```
# 2. Bar Chart (Daily % Change of Stock A)
```

```
change = [0]
```

```
for i in range(1, len(a)):
```

```
    p = ((a[i] - a[i-1]) / a[i-1]) * 100
```

```
    change.append(round(p, 2))
```

```
plt.bar(date, change)
plt.title("Daily % Change - Stock A")
plt.xlabel("Date")
plt.ylabel("Percentage Change")
plt.show()
```

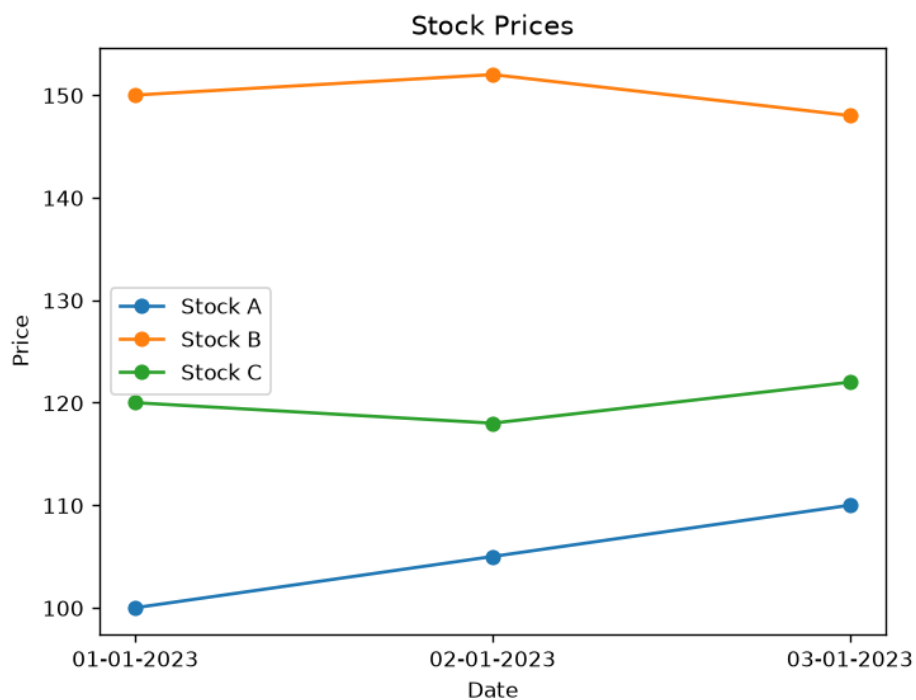
3. Table

```
print("Date\t\tStock A\tStock B\tStock C")
for i in range(len(date)):
    print(date[i], "\t", a[i], "\t", b[i], "\t", c[i])
```

4. Tableau Dashboard

```
print("\nImport Stock_Prices.csv into Tableau")
print("Create Line Chart, Bar Chart and Table")
print("Combine them into one Dashboard")
```

Output:



Set 21

Sector	Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost (₹)	Renewable_Usage (%)	Peak_Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

1. Import the dataset **in R**. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

R program:

```
# Load required package
```

```
library(ggplot2)
```

```
# Create the dataset
```

```
energy_data <- data.frame(
```

```
  Sector = c("Residential", "Commercial", "Industrial",
```

```
            "Residential", "Commercial", "Industrial"),
```

```
  Region = c("North", "South", "West", "East", "North", "South"),
```

```
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
```

```
  Temperature = c(15, 24, 20, 18, 28, 30),
```

```
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
```

```
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
```

```
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
```

```
Peak_Hours = c(4, 6, 8, 5, 7, 9)

)

# Histogram

ggplot(energy_data, aes(x = Units_Consumed)) +

  geom_histogram(

    bins = 5,

    fill = "skyblue",

    color = "black"

  ) +

  labs(

    title = "Histogram of Energy Consumption",

    x = "Units Consumed (kWh)",

    y = "Frequency"

  ) +

  theme_minimal()
```

```
# Density Plot

ggplot(energy_data, aes(x = Units_Consumed)) +

  geom_density(

    fill = "lightgreen",

    alpha = 0.5

  ) +

  labs(
```



```

title = "Density Plot of Energy Consumption",

x = "Units Consumed (kWh)",

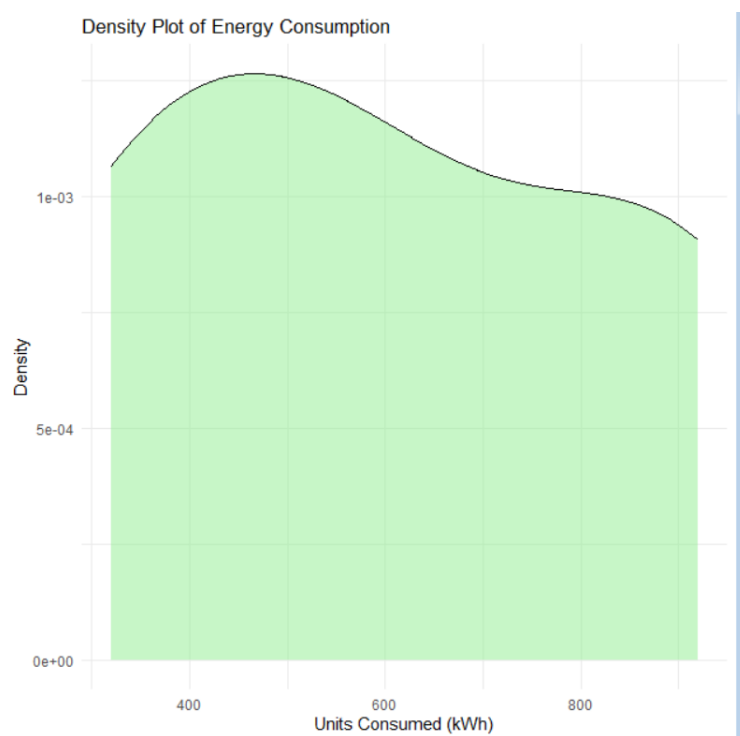
y = "Density"

) +

theme_minimal()

```

Output:



- Using **R programming**, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

R program:

```

library(ggplot2)

ggplot(energy_data,
  aes(x = Temperature,
    y = Units_Consumed,
    size = Peak_Hours)) +
  geom_point(
    alpha = 0.6,

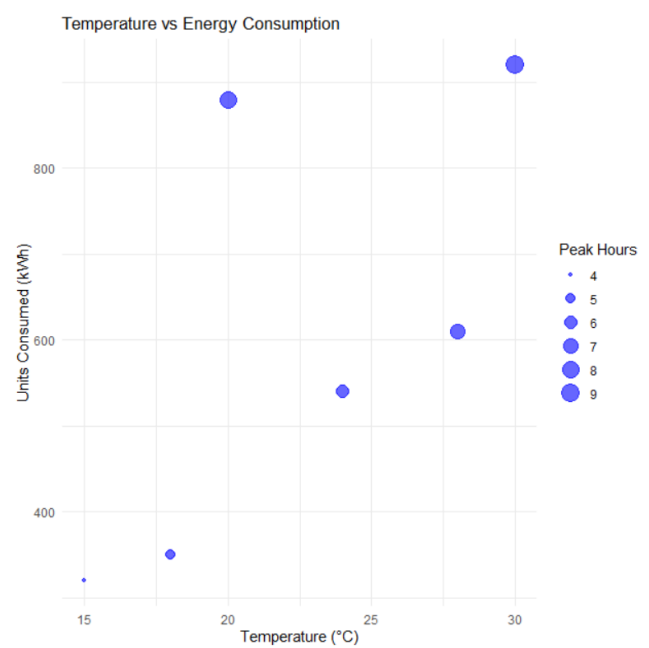
```

```

    color = "blue"
  ) +
  labs(
    title = "Temperature vs Energy Consumption",
    x = "Temperature (°C)",
    y = "Units Consumed (kWh)",
    size = "Peak Hours"
  ) +
  theme_minimal()

```

Output:



- Using **R**, calculate average Renewable_Usage for each Sector. Create a bar chart.
Interpret renewable energy adoption.

R program:

```

library(ggplot2)

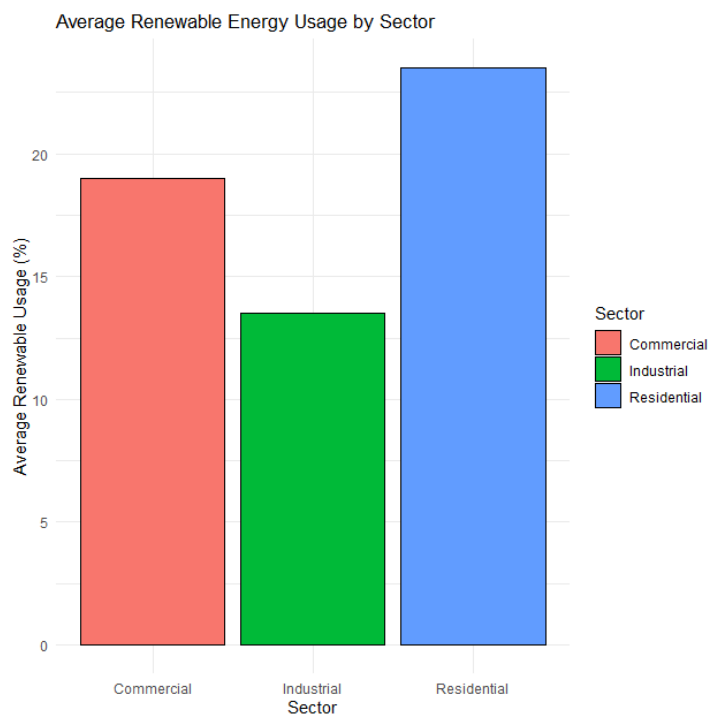
# Calculate average renewable usage
avg_renewable <- aggregate(
  Renewable_Usage ~ Sector,
  data = energy_data,
  FUN = mean
)

```

```
# Display the result
print(avg_renewable)

# Create bar chart
ggplot(avg_renewable,
       aes(x = Sector,
           y = Renewable_Usage,
           fill = Sector)) +
  geom_bar(
    stat = "identity",
    color = "black"
  ) +
  labs(
    title = "Average Renewable Energy Usage by Sector",
    x = "Sector",
    y = "Average Renewable Usage (%)"
  ) +
  theme_minimal()
```

Output:



Correct Result

Sector	Average Renewable Usage (%)
Commercial	19.0
Industrial	13.5
Residential	23.5

Calculation

Residential:

$$222+25=23.5\%$$

Commercial:

$$218+20=19\%$$

Industrial:

$$212+15=13.5\%$$

Interpretation

The **Residential sector has the highest renewable-energy adoption at 23.5%**, followed by Commercial at 19%. The **Industrial sector has the lowest adoption at 13.5%**. Therefore, the industrial sector has the greatest scope for increasing renewable-energy usage.

4. Import Energy_Consumption_Data.csv into **Tableau**. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.

Ans:

Step 1: Import the CSV

Open Tableau and select:

Connect → To a File → Text File

Select:

Energy_Consumption_Data.csv

Make sure Tableau recognizes these fields:

Sector

Region

Month

Temperature

Units_Consumed

Cost

Renewable_Usage

Peak_Hours

A. Monthly Energy Usage Line Chart

Create a worksheet named:

Monthly Energy Usage

Set:

- **Columns → Month**
- **Rows → SUM(Units_Consumed)**
- **Marks → Line**
- **Color → Sector**

The chart displays the monthly energy consumption trend for different sectors.

For the given data:

Month Total Units Consumed

Jan 860

Feb 1230

Month Total Units Consumed

Mar 1530

So consumption increases from **January to March**.

B. Sector vs Region Heatmap

Create another worksheet named:

Sector-Region Heatmap

Set:

- Rows → Sector
- Columns → Region
- Marks → Square
- Color → SUM(Units_Consumed)
- Label → SUM(Units_Consumed)

The heatmap will show energy consumption using different color intensities.

For example:

Sector	Region	Units
Residential	North	320
Residential	East	350
Commercial	South	540
Commercial	North	610
Industrial	West	880
Industrial	South	920

Higher values will appear with a stronger color intensity.

C. Add Month Filter

In the worksheet or dashboard:

1. Drag **Month** to the **Filters** shelf.
2. Right-click `Month`.
3. Select **Show Filter**.

Now users can select **Jan, Feb, or Mar**.

D. Add Sector Filter

1. Drag **Sector** to the **Filters** shelf.
2. Select **Show Filter**.

Users can now select:

- Residential
- Commercial
- Industrial

E. Create the Interactive Dashboard

Create:

Dashboard → New Dashboard

Add these two worksheets:

1. **Monthly Energy Usage**
2. **Sector-Region Heatmap**

Add the filters:

- **Month**
- **Sector**

Final Dashboard Layout

ENERGY CONSUMPTION ANALYSIS	
DASHBOARD	
Month Filter	Sector Filter
MONTHLY ENERGY USAGE	
LINE CHART	
SECTOR vs REGION HEATMAP	

Final Interpretation

The Tableau dashboard provides an interactive view of energy consumption. The line chart shows that total energy usage increases from **860 kWh in January to 1,530 kWh in March**. The heatmap allows comparison of consumption across sectors and regions. The **Month and Sector filters** allow users to dynamically explore specific portions of the dataset.

Set 22: Time Series Analysis

Dataset: Monthly Sales Data

Month	Sales (in \$)
January	15000
February	18000
March	22000
April	20000
May	23000

1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

R program:

Enter the data


```
Month <- c("January", "February", "March", "April", "May")
```

```
Sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
# Create data frame
```

```
sales_data <- data.frame(Month, Sales)
```

```
# Create time series
```

```
sales_ts <- ts(
```

```
  Sales,
```

```
  start = c(2026, 1),
```

```
  frequency = 12
```

```
)
```

```
# Plot time series
```

```
plot(
```

```
  sales_ts,
```

```
  type = "o",
```

```
  col = "blue",
```

```
  lwd = 2,
```

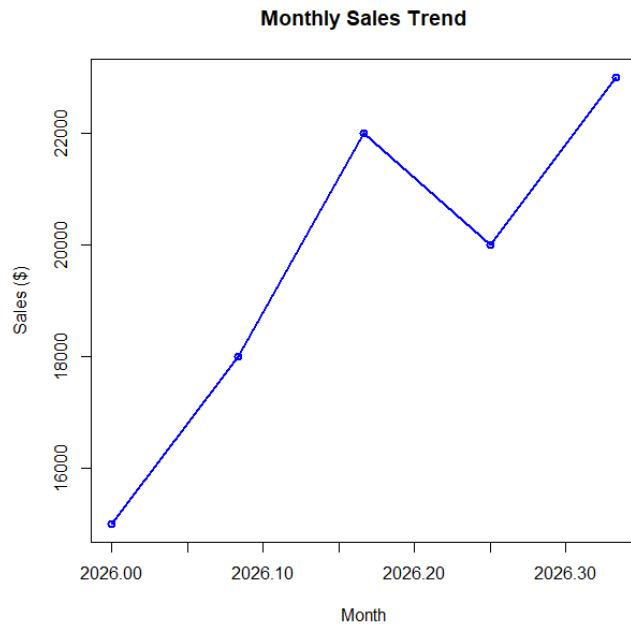
```
  xlab = "Month",
```

```
  ylab = "Sales ($)",
```

```
  main = "Monthly Sales Trend"
```

```
)
```

Output:



2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

R program:

Advertising budget data

```
Advertising_Budget <- c(3000, 4000, 5000, 4500, 6000)
```

Create data frame

```
ad_data <- data.frame(
```

```
  Month = Month,
```

```
  Advertising_Budget = Advertising_Budget,
```

```
  Sales = Sales
```

```
)
```

Scatter plot

```
plot(
```

```
  ad_data$Advertising_Budget,
```

```
  ad_data$Sales,
```

```
  pch = 19,
```

```
  col = "darkgreen",
```

```
  xlab = "Advertising Budget ($)",
```

```
  ylab = "Monthly Sales ($)",
```

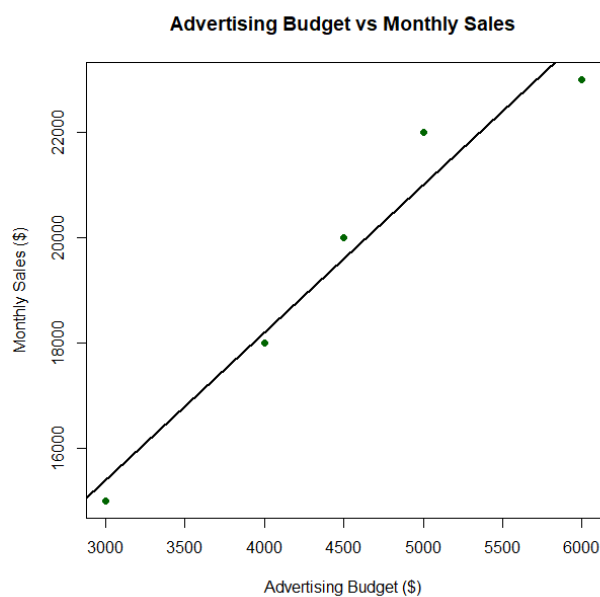
```

main = "Advertising Budget vs Monthly Sales"
)

# Add trend line
abline(
  lm(Sales ~ Advertising_Budget, data = ad_data),
  lwd = 2
)

```

Output:



3. In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.

Ans:

```

Month, Product, Sales
January, Product A, 5000
January, Product B, 4000
January, Product C, 6000
February, Product A, 6000
February, Product B, 5000
February, Product C, 7000
March, Product A, 7000
March, Product B, 6000
March, Product C, 9000
April, Product A, 6500
April, Product B, 5500

```

April, Product C, 8000
May, Product A, 7500
May, Product B, 6500
May, Product C, 9000

Tableau Line Chart

1. Open **Tableau**.
2. Select **Connect** → **Text File**.
3. Import the CSV file.
4. Create a new worksheet.
5. Drag **Month** → Columns.
6. Drag **Sales** → Rows.
7. Select **Line** from Marks.
8. Drag **Product** → Color.

This creates the **Monthly Sales Trend**.

Tableau Pie Chart

1. Create another worksheet.
2. Drag **Product** → Color.
3. Drag **Sales** → Angle.
4. Drag **Sales** → Label.
5. Select **Pie** from the Marks card.

This shows the **distribution of sales across products**.

Create Dashboard

Go to:

Dashboard → **New Dashboard**

Add:

- Monthly Sales Trend

- Product Sales Distribution

Suggested layout:

MONTHLY SALES DASHBOARD	
SALES TREND LINE CHART	PRODUCT SALES PIE CHART

4. In R, create an autocorrelation plot to identify seasonality in the time series data.

R program:

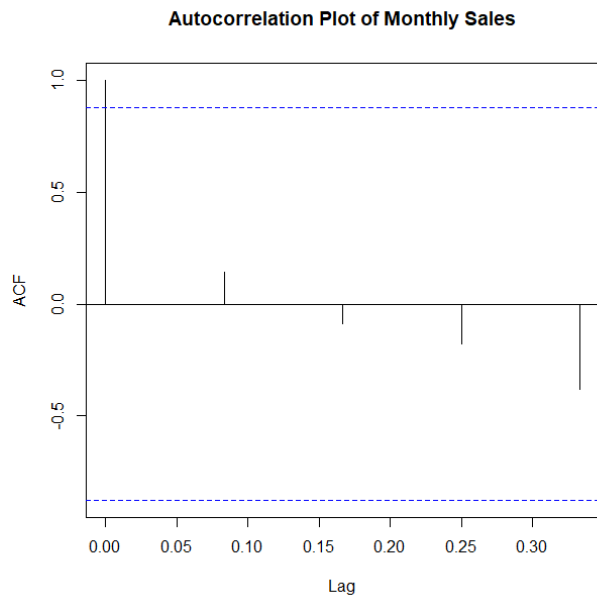
Create time series

```
sales_ts <- ts(
  Sales,
  start = c(2026, 1),
  frequency = 12
)
```

Autocorrelation plot

```
acf(
  sales_ts,
  main = "Autocorrelation Plot of Monthly Sales",
  lag.max = 4
)
```

Output:



Set 23: Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

Ans:

Employee data

```
employee_data <- data.frame(
```

```
  Employee_ID = c(1, 2, 3, 4, 5),
```

```
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
```

```
  Years_of_Service = c(5, 3, 7, 4, 2),
```

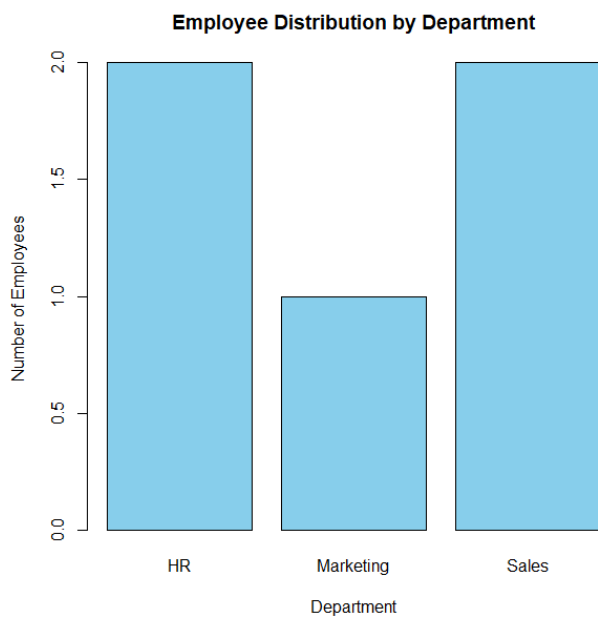
```
  Performance_Score = c(85, 92, 78, 90, 76)
```

```
)
```

```
# Count employees in each department
department_count <- table(employee_data$Department)
```

```
# Create bar chart
barplot(
  department_count,
  main = "Employee Distribution by Department",
  xlab = "Department",
  ylab = "Number of Employees",
  col = "skyblue",
  border = "black"
)
```

Output:



2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

R program:

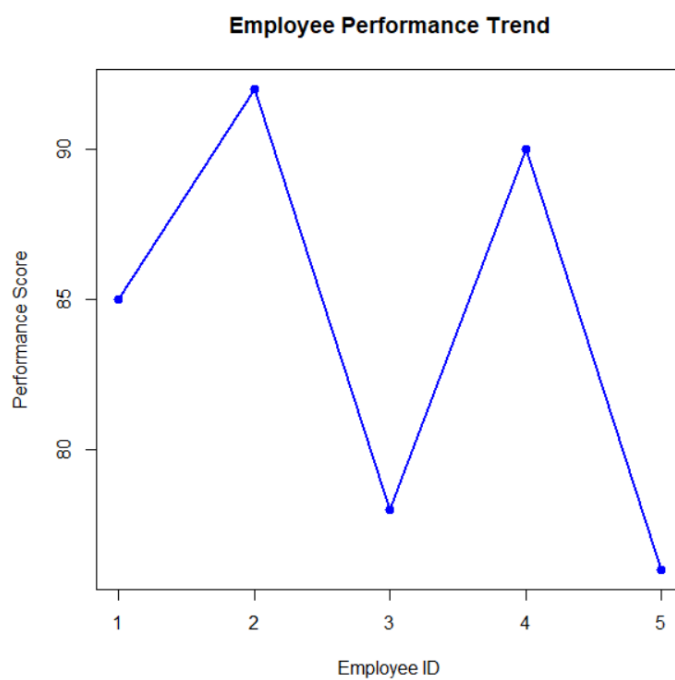
```
# Employee data
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
```

```
Performance_Score = c(85, 92, 78, 90, 76)  
)
```

```
# Create line chart
```

```
plot(  
  employee_data$Employee_ID,  
  employee_data$Performance_Score,  
  type = "o",  
  pch = 19,  
  col = "blue",  
  lwd = 2,  
  main = "Employee Performance Trend",  
  xlab = "Employee ID",  
  ylab = "Performance Score"  
)
```

Output:



3. In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.

Ans:

Employee_ID,Department,Years_of_Service,Performance_Score

1,Sales,5,85

2,HR,3,92

3,Marketing,7,78

4,Sales,4,90

5,HR,2,76

Sheet 1 – Department Bar Chart

1. Open Tableau.
2. Select **Connect** → **Text File**.
3. Import Employee_Performance.csv.
4. Create a new worksheet.
5. Drag **Department** → Columns.
6. Drag **Employee_ID** → Rows.
7. Change it to **COUNT(Employee_ID)**.
8. Select **Bar** from Marks.

Title: Employee Distribution by Department

Sheet 2 – Scatter Plot

1. Create a new worksheet.
2. Drag **Years_of_Service** → Columns.
3. Drag **Performance_Score** → Rows.
4. Select **Circle** from Marks.
5. Drag **Department** → Color.
6. Drag **Employee_ID** → Label.

Title: Years of Service vs Performance Score

Dashboard

Go to:

Dashboard → New Dashboard

Add both worksheets.

EMPLOYEE PERFORMANCE DASHBOARD	
DEPARTMENT DISTRIBUTION BAR CHART	YEARS OF SERVICE VS PERFORMANCE SCATTER PLOT

4. In R, develop a table showing the performance data for each employee. Label the table elements.

R program:

Employee data

```
employee_data <- data.frame(  
  Employee_ID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  Years_of_Service = c(5, 3, 7, 4, 2),  
  Performance_Score = c(85, 92, 78, 90, 76)  
)
```

Display the table

```
print(employee_data)
```

Open as a table in RStudio

```
View(employee_data)
```

Output:

	Employee_ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78
4	4	Sales	4	90
5	5	HR	2	76

Set 24 Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

R program:

Create the dataset

```
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C",
    "Product D", "Product E"),
  Quantity_Available = c(250, 175, 300, 200, 220)
)
```

Create bar chart

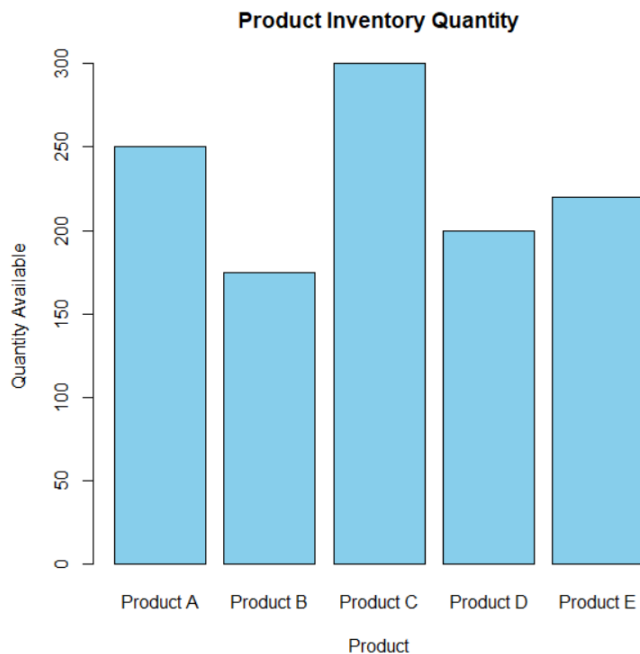
```
barplot(
  inventory_data$Quantity_Available,
  names.arg = inventory_data$Product_Name,
  main = "Product Inventory Quantity",
```

```

xlab = "Product",
ylab = "Quantity Available",
col = "skyblue",
border = "black"
)

```

Output:



2. In R, generate a stacked bar chart to show the quantity of each product within different product categories.

R program:

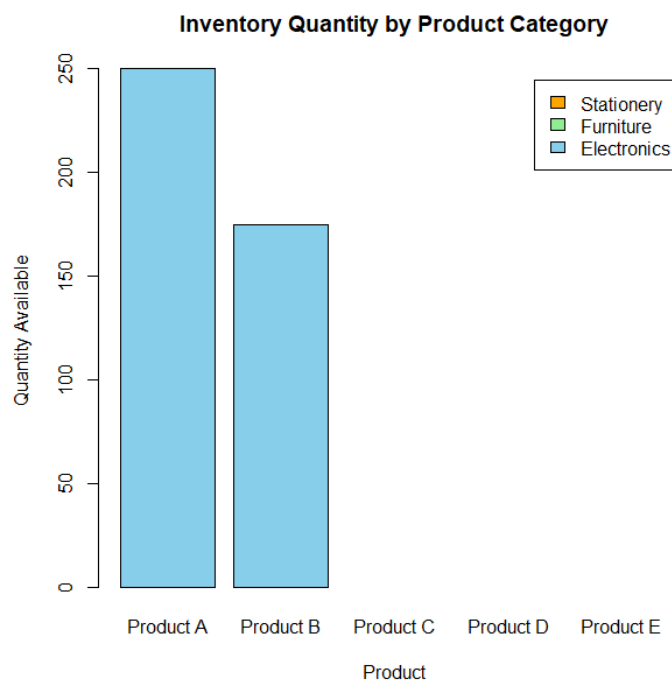
```

# Create dataset with categories
inventory_category <- data.frame(
  Product_Name = c(
    "Product A", "Product B", "Product C",
    "Product D", "Product E"
  ),
  Category = c(
    "Electronics", "Electronics", "Furniture",
    "Furniture", "Stationery"
  ),
  Quantity_Available = c(250, 175, 300, 200, 220)
)

```

```
# Create stacked bar chart
barplot(
  tapply(
    inventory_category$Quantity_Available,
    list(
      inventory_category$Category,
      inventory_category$Product_Name
    ),
    sum
  ),
  main = "Inventory Quantity by Product Category",
  xlab = "Product",
  ylab = "Quantity Available",
  col = c("skyblue", "lightgreen", "orange"),
  legend.text = TRUE,
  border = "black"
)
```

Output:



3. In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.

Ans:

Use a CSV file such as:

```
Product_ID,Product_Name,Category,Quantity_Available
1,Product A,Electronics,250
2,Product B,Electronics,175
3,Product C,Furniture,300
4,Product D,Furniture,200
5,Product E,Stationery,220
```

Save it as:

Product_Inventory.csv

Sheet 1 – Bar Chart

In Tableau:

1. Connect to Product_Inventory.csv.
2. Create a new worksheet.
3. Drag **Product Name** → Columns.
4. Drag **Quantity Available** → Rows.
5. Select **Bar** from Marks.
6. Add **Quantity Available** → Label.

Title:

Product Inventory Quantity

Sheet 2 – Stacked Bar Chart

1. Create another worksheet.
2. Drag **Category** → Columns.
3. Drag **Quantity Available** → Rows.
4. Drag **Product Name** → Color.
5. Select **Bar**.

Title:

Inventory Quantity by Product Category

Create Dashboard

Go to:

Dashboard → New Dashboard

Add both worksheets.

PRODUCT INVENTORY DASHBOARD	
PRODUCT QUANTITY BAR CHART	CATEGORY-WISE STACKED BAR CHART

4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

R program:

```
# Create dataset with product price
```

```
price_data <- data.frame(  
  Product_Name = c(  
    "Product A", "Product B", "Product C",  
    "Product D", "Product E"  
  ),  
  Price = c(500, 750, 400, 600, 300),  
  Quantity_Available = c(250, 175, 300, 200, 220)  
)
```

```
# Create scatter plot
```

```
plot(  
  price_data$Price,  
  price_data$Quantity_Available,  
  pch = 19,  
  col = "blue",  
  main = "Product Price vs Quantity Available",  
  xlab = "Product Price ($)",  
  ylab = "Quantity Available"  
)
```

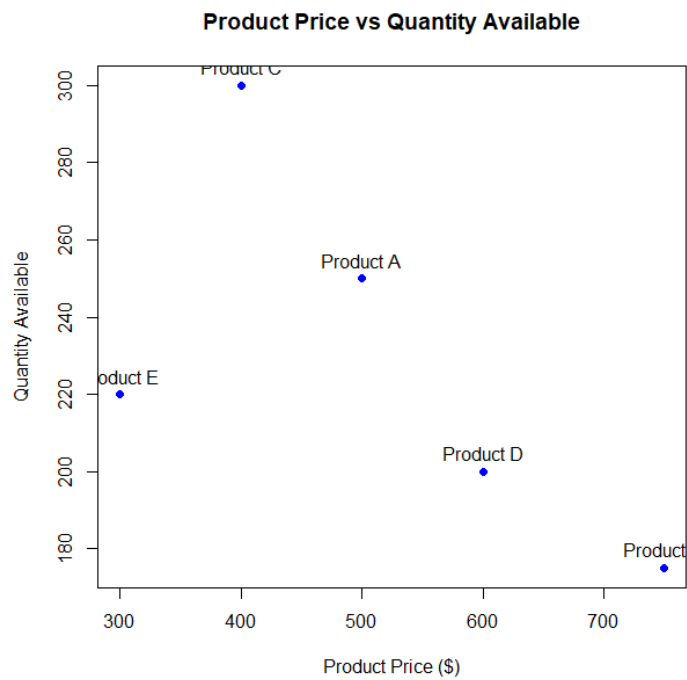
```
# Add product labels
```

```

text(
  price_data$Price,
  price_data$Quantity_Available,
  labels = price_data$Product_Name,
  pos = 3
)

```

Output:



Set 25: Website Traffic Analysis

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

1. In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.

Ans:

Dataset

Date,Page_Views,Click_Through_Rate

2023-01-01,1500,2.3

2023-01-02,1600,2.7

2023-01-03,1400,2.0

2023-01-04,1650,2.4

2023-01-05,1800,2.6

Save as:

Website_Traffic.csv

Tableau Steps

1. Open **Tableau**.
2. Select **Connect** → **Text File**.
3. Import Website_Traffic.csv.
4. Create a new worksheet.
5. Drag **Date** → Columns.
6. Drag **Page Views** → Rows.
7. Select **Line** from the Marks card.

Title

Website Page Views Trend

Axis Labels

- X-axis: **Date**
- Y-axis: **Page Views**

Interpretation

Page views fluctuate during the five-day period. The highest page views occur on **January 5 with 1,800 views**, while the lowest occur on **January 3 with 1,400 views**.

2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

R program:

```
# Create the dataset
```

```
traffic_data <- data.frame(
```

```
  Date = as.Date(c(
```

```
    "2023-01-01",
```

```
    "2023-01-02",
```

```
    "2023-01-03",
```

```
    "2023-01-04",
```

```
    "2023-01-05"
```

```
  )),
```

```
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
```

```
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)
```

```
)
```

```
# Set Top N
```

```
N <- 3
```

```
# Sort by click-through rate
```

```
top_days <- traffic_data[  
  
  order(-traffic_data$Click_Through_Rate),  
  
]
```

```
# Select Top N
```

```
top_days <- head(top_days, N)
```

```
# Display Top N
```

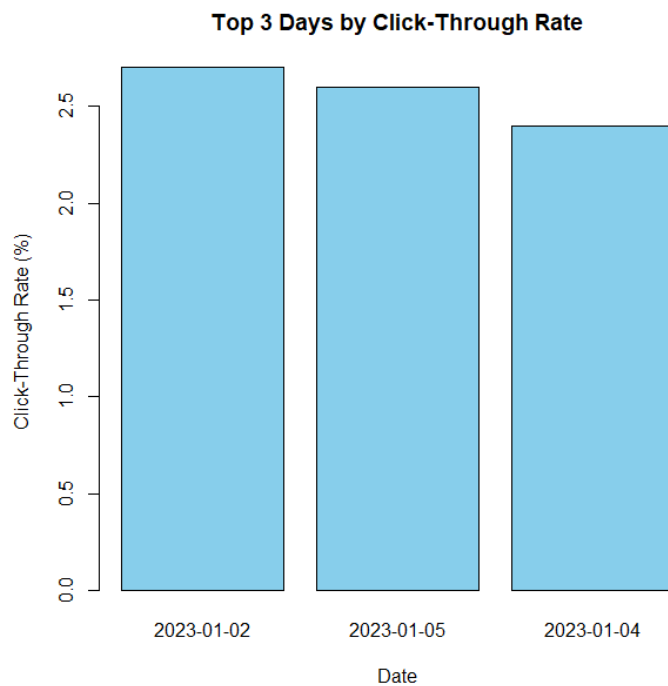
```
print(top_days)
```

```
# Create bar chart
```

```
barplot(  
  
  top_days$Click_Through_Rate,  
  
  names.arg = format(top_days$Date, "%Y-%m-%d"),  
  
  main = "Top 3 Days by Click-Through Rate",  
  
  xlab = "Date",  
  
  ylab = "Click-Through Rate (%)",  
  
  col = "skyblue",  
  
  border = "black"
```

)

Output:



3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

R program:

```
# Create user interaction data
```

```
interaction_data <- data.frame(
```

```
  Date = as.Date(c(
```

```
    "2023-01-01",
```

```
    "2023-01-02",
```

```
    "2023-01-03",
```

```
    "2023-01-04",
```

```
    "2023-01-05"
```

```
  )),
```

```
Likes = c(120, 150, 110, 170, 200),  
  
Shares = c(45, 55, 40, 60, 75),  
  
Comments = c(20, 25, 18, 30, 35)  
  
)
```

```
# Create stacked area chart
```

```
plot(  
  
  interaction_data$Date,  
  
  interaction_data$Likes,  
  
  type = "s",  
  
  lwd = 2,  
  
  col = "blue",  
  
  xlab = "Date",  
  
  ylab = "Number of Interactions",  
  
  main = "User Interactions Over Time"  
  
)
```

```
# Add Shares
```

```
lines(  
  
  interaction_data$Date,
```

```
interaction_data$Likes +  
  
  interaction_data$Shares,  
  
type = "s",  
  
lwd = 2,  
  
col = "green"  
)
```

```
# Add Comments
```

```
lines(  
  
  interaction_data$Date,  
  
  interaction_data$Likes +  
  
    interaction_data$Shares +  
  
    interaction_data$Comments,  
  
type = "s",  
  
lwd = 2,  
  
col = "red"  
)
```

```
# Add legend
```

```
legend(  

```

"topleft",

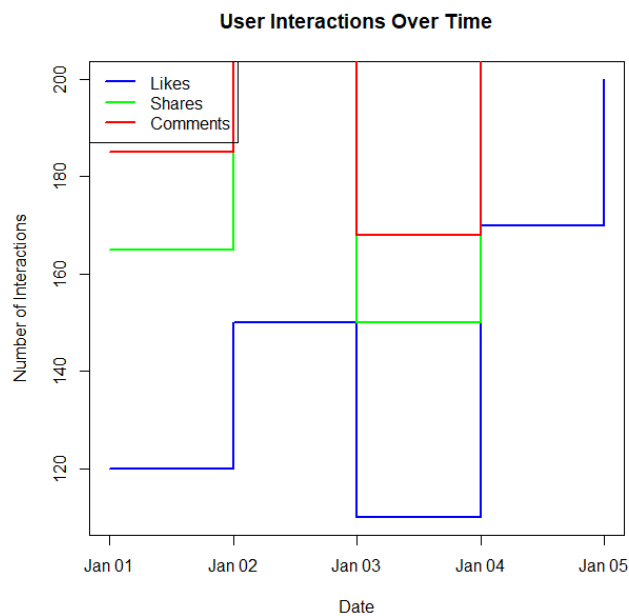
legend = c("Likes", "Shares", "Comments"),

col = c("blue", "green", "red"),

lwd = 2

)

Output:



4. In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

Ans:

A. Create the Map

1. Import the CSV into Tableau.
2. Create a new worksheet.
3. Double-click **Country**.
4. Tableau automatically generates a map.
5. Drag **Page Views** → Color.
6. Drag **Page Views** → Size.
7. Drag **Traffic Source** → Detail.

Title:

Website Traffic Sources by Country

B. Create Page Views by Source Bar Chart

Create another worksheet.

Drag:

Traffic Source → Columns

Page Views → Rows

Select **Bar** from Marks.

Title:

Page Views by Traffic Source

C. Create Interactive Dashboard

Go to:

Dashboard → **New Dashboard**

Add:

- **Traffic Sources Map**
- **Page Views by Traffic Source**

WEBSITE TRAFFIC DASHBOARD	
TRAFFIC SOURCE MAP	PAGE VIEWS BY TRAFFIC SOURCE BAR CHART

You can add **Traffic Source** as a filter to make the dashboard interactive.

Final Interpretation

The Tableau dashboard allows users to identify **where website visitors come from geographically** and compare the **number of page views generated by each traffic source**. Selecting a traffic source can be used to interactively explore its corresponding traffic on the map and bar chart.

SET 26

Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

1. Import the dataset **in R**. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

R program:

```
# Create the dataset
```

```
student_data <- data.frame(
```

```
  Student_ID = c("S01", "S02", "S03", "S04", "S05", "S06"),
```

```
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
```

```
  Study_Hours = c(2.0, 3.5, 1.5, 4.0, 2.8, 3.0),
```

```
  Attendance = c(78, 90, 70, 95, 85, 92),
```

```
  Math_Score = c(62, 80, 55, 90, 72, 82),
```

```
  Science_Score = c(65, 85, 58, 92, 74, 86),
```

```
  Exam_Date = as.Date(c(
```

```
    "2025-01-10",
```

```
    "2025-01-10",
```

```
    "2025-02-12",
```

```
    "2025-02-12",
```

```
    "2025-03-15",
```

```
    "2025-03-15"
```

```

))
)

# Display dataset
print(student_data)

# -----
# Histogram of Math Score
# -----

hist(
  student_data$Math_Score,
  main = "Distribution of Math Scores",
  xlab = "Math Score",
  ylab = "Frequency",
  col = "skyblue",
  border = "black"
)

# -----
# Boxplot of Science Score by Gender
# -----

boxplot(
  Science_Score ~ Gender,
  data = student_data,
  main = "Science Scores by Gender",
  xlab = "Gender",
  ylab = "Science Score",
  col = c("lightblue", "pink")
)
Output:

```



2. Create a scatter plot **using R programming** between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

R program:

Create scatter plot

```
plot(
  student_data$Study_Hours,
  student_data$Math_Score,
  pch = 19,
  col = ifelse(student_data$Gender == "Male", "blue", "red"),
  main = "Study Hours vs Math Score",
  xlab = "Study Hours",
  ylab = "Math Score"
)
```

Add regression line

```
model <- lm(
  Math_Score ~ Study_Hours,
  data = student_data
)
```

```
abline(
  model,
```

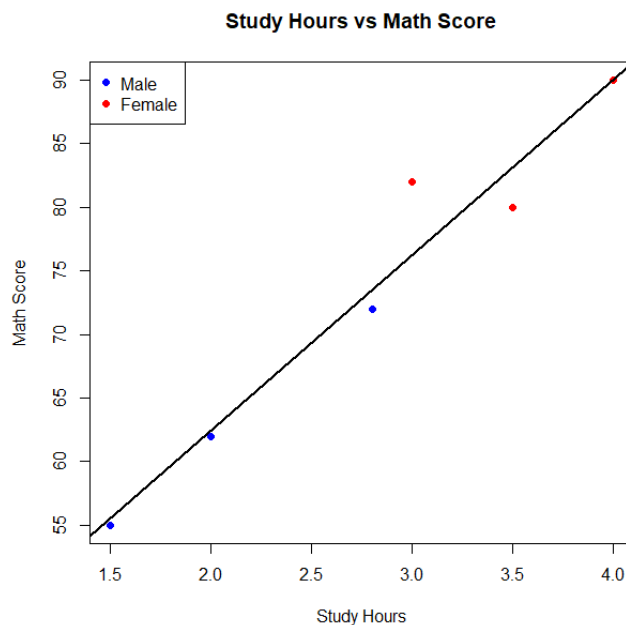
```

col = "black",
lwd = 2
)

# Add legend
legend(
  "topleft",
  legend = c("Male", "Female"),
  col = c("blue", "red"),
  pch = 19
)

```

Output:



- Using **R**, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

R program:

```

# Convert Exam_Date to Date format
student_data$Exam_Date <- as.Date(
  student_data$Exam_Date,
  format = "%Y-%m-%d"
)

```

```

# Create Month column

```

```

student_data$Month <- format(
  student_data$Exam_Date,
  "%Y-%m"
)

# Calculate monthly average Math score
monthly_avg <- aggregate(
  Math_Score ~ Month,
  data = student_data,
  FUN = mean
)

# Display monthly averages
print(monthly_avg)

# -----
# Calculate 2-month moving average
# -----

monthly_avg$Moving_Average <- c(
  NA,
  (monthly_avg$Math_Score[1] +
   monthly_avg$Math_Score[2]) / 2,
  (monthly_avg$Math_Score[2] +
   monthly_avg$Math_Score[3]) / 2
)

# Display result
print(monthly_avg)

# -----
# Line chart
# -----

plot(

```

```
1:nrow(monthly_avg),
monthly_avg$Math_Score,
type = "o",
pch = 19,
col = "blue",
lwd = 2,
xaxt = "n",
main = "Monthly Average Math Score Trend",
xlab = "Month",
ylab = "Average Math Score"
)
```

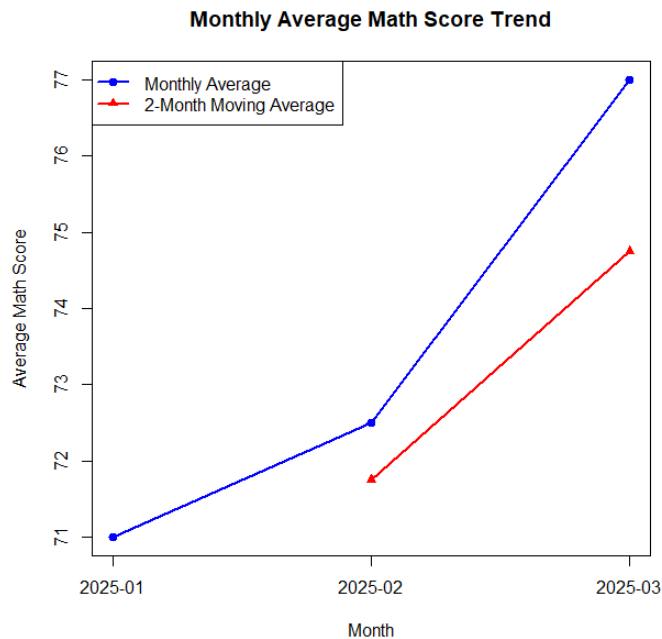
```
# Add month labels
axis(
  1,
  at = 1:nrow(monthly_avg),
  labels = monthly_avg$Month
)
```

```
# Add moving average line
lines(
  1:nrow(monthly_avg),
  monthly_avg$Moving_Average,
  type = "o",
  pch = 17,
  col = "red",
  lwd = 2
)
```

```
# Add legend
legend(
  "topleft",
  legend = c(
    "Monthly Average",
    "2-Month Moving Average"
  ),
  col = c("blue", "red"),
```

```
pch = c(19, 17),
lwd = 2
)
```

Output:



4. Import Student_Mini_Data.csv into **Tableau**. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for Exam_Date. Design a simple interactive dashboard.

Ans:

Sheet 1 – Average Math Score by Gender

1. Open **Tableau**.
2. Select **Connect** → **Text File**.
3. Import Student_Mini_Data.csv.
4. Create a new worksheet.
5. Drag **Gender** → Columns.
6. Drag **Math_Score** → Rows.
7. Make sure it is **AVG(Math_Score)**.
8. Select **Bar** from Marks.

Title:

Average Math Score by Gender

Expected averages:

Gender Average Math Score

Female 84.0

Male 63.0

Sheet 2 – Score Trend Over Time

1. Create a new worksheet.
2. Drag **Exam_Date** → Columns.
3. Drag **Math_Score** → Rows.
4. Change aggregation to **AVG**.
5. Select **Line** from Marks.

Title:

Math Score Trend Over Time

Add Exam Date Filter

On the line-chart worksheet:

1. Drag **Exam_Date** → Filters.
2. Select the required date range.
3. Right-click Exam_Date.
4. Select **Show Filter**.

This allows the user to select specific examination dates.

Tableau Dashboard

Go to:

Dashboard → New Dashboard

Add:

- **Average Math Score by Gender**
- **Math Score Trend Over Time**
- **Exam Date Filter**

Suggested Layout

STUDENT PERFORMANCE DASHBOARD	
Exam Date Filter	
AVERAGE MATH SCORE BY GENDER BAR CHART	MATH SCORE TREND OVER TIME LINE CHART

Final Interpretation

The dashboard shows that the **average Math score of female students is 84**, compared with **63 for male students** in this dataset. The monthly trend also shows an improvement in Math performance from January to March. The Exam Date filter allows users to interactively examine scores for specific examination dates.

SET 27.

Patient Health Risk Analysis

Patient_ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

R program:

```
# Create the dataset
```

```
patient_data <- data.frame(
```

```
  Patient_ID = c("P1", "P2", "P3", "P4", "P5"),
```

```
  Age = c(25, 40, 55, 35, 60),
```

```
  BMI = c(22, 28, 30, 26, 32),
```

```
  BP = c(120, 135, 145, 130, 150),
```

```
  Cholesterol = c(180, 210, 240, 200, 260)
```

```
)
```

```
# Create scatterplot matrix
```

```
pairs(
```

```
  patient_data[, c("Age", "BMI", "BP", "Cholesterol")],
```

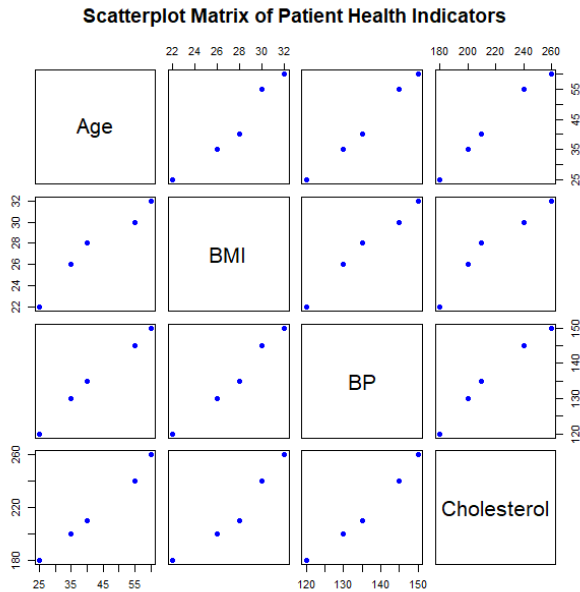
```
  main = "Scatterplot Matrix of Patient Health Indicators",
```

```
  pch = 19,
```

```
  col = "blue"
```

```
)
```

Output:



2. Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

R program:

```
# Cholesterol data
```

```
cholesterol <- patient_data$Cholesterol
```

```
# -----
```

```
# Q-Q Plot
```

```
# -----
```

```
qqnorm(
```

```
cholesterol,
```

```
main = "Q-Q Plot of Cholesterol Levels",
```

```
pch = 19,
```

```
col = "blue"

)

qqline(

cholesterol,

col = "red",

lwd = 2

)

# -----

# ECDF

# -----

plot(

ecdf(cholesterol),

main = "ECDF of Cholesterol Levels",

xlab = "Cholesterol Level",

ylab = "Cumulative Probability",

verticals = TRUE,
```

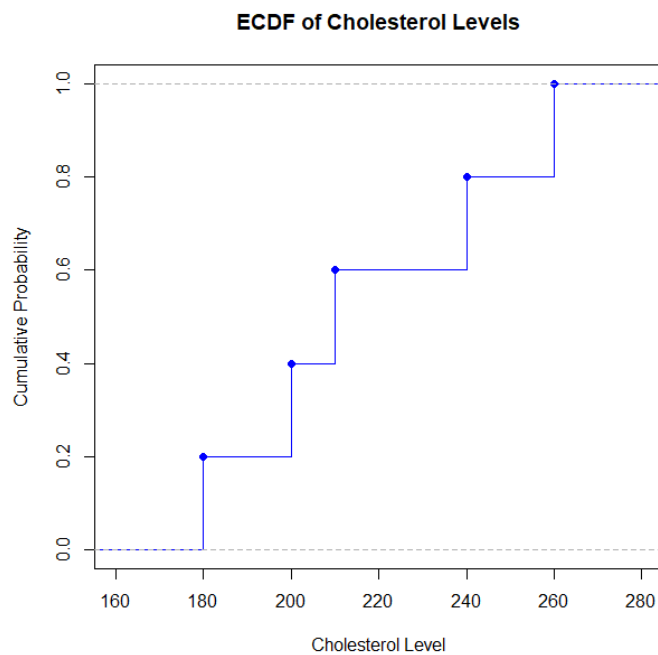
```
do.points = TRUE,

pch = 19,

col = "blue"

)
```

Output:



3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk

R program:

```
# Calculate average values

average_values <- c(

  mean(patient_data$Age),

  mean(patient_data$BMI),

  mean(patient_data$BP),
```

```
mean(patient_data$Cholesterol)

)

# Indicator names

indicators <- c(

  "Age",

  "BMI",

  "BP",

  "Cholesterol"

)

# Create bar chart

barplot(

  average_values,

  names.arg = indicators,

  col = c("skyblue", "lightgreen", "orange", "pink"),

  border = "black",

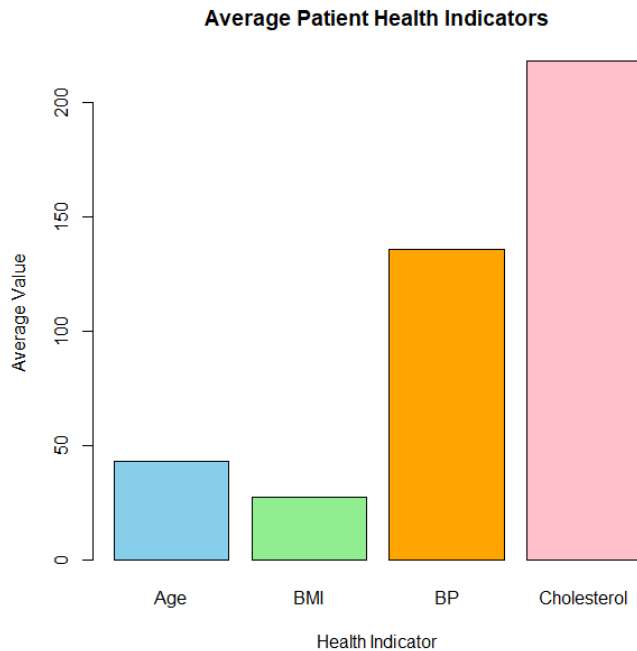
  main = "Average Patient Health Indicators",

  xlab = "Health Indicator",

  ylab = "Average Value"
```

)

Output:



4. Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

Ans:

Step 1 – Import Dataset

Open Tableau:

Connect → Text File → Patient_Health_Risk.csv

Step 2 – Create Heatmap

Create a new worksheet named:

Patient Risk Heatmap

Set:

Rows → Patient_ID

Columns → Measure Names

Marks → Square

Then:

Measure Values → Color

Measure Values → Label

Use:

- Age
- BMI
- BP
- Cholesterol

for the health indicators.

A simpler and clearer alternative is to use:

Rows → Patient_ID

Columns → Age

Color → Cholesterol

and add BMI and BP to the tooltip.

Tooltip

Drag the following fields to **Tooltip**:

Age

BMI

BP

Cholesterol

So when the user hovers over a patient, all health indicators are displayed.

Step 3 – Add Age Filter

Drag:

Age → Filters

Then select:

Show Filter

Users can now select an age range to explore different patient groups.

Step 4 – Create Dashboard

Go to:

Dashboard → New Dashboard

Add the heatmap and Age filter.

Suggested Layout

PATIENT HEALTH RISK DASHBOARD				
Age Filter: [-----]				
PATIENT RISK HEATMAP				
Patient	Age	BMI	BP	Cholesterol
P1	■	■	■	■
P2	■	■	■	■
P3	■	■	■	■
P4	■	■	■	■
P5	■	■	■	■

Risk Analysis

Based on the **values in this small dataset**, **P5** has the highest combination of age, BMI, BP, and cholesterol:

Patient Age BMI BP Cholesterol

Patient Age BMI BP Cholesterol

P5 60 32 150 260

P3 55 30 145 240

P2 40 28 135 210

P4 35 26 130 200

P1 25 22 120 180

There is also an apparent pattern in this sample: **age increases along with BMI, BP, and cholesterol**. P1 has the lowest values, while P5 has the highest values.

Dashboard Use

The interactive dashboard allows healthcare professionals to:

- Compare health indicators between patients.
- Filter patients by **age**.
- Quickly identify patients with relatively high indicator values.
- Hover over a patient to see **Age, BMI, BP, and Cholesterol**.
- Prioritize which records may warrant closer clinical review.

SET 28. Vehicle Performance Analysis

Dataset:

Vehicle_ID	Engine_Size (L)	Horsepower	Fuel_Efficiency (km/l)	Top_Speed (km/h)	Safety_Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

R program:

```
# Create the dataset
```

```
vehicle_data <- data.frame(  
  
  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),  
  
  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),  
  
  Horsepower = c(110, 150, 250, 200, 130),  
  
  Fuel_Efficiency = c(18, 15, 12, 14, 17),  
  
  Top_Speed = c(180, 200, 250, 220, 190),  
  
  Safety_Rating = c(4, 5, 5, 4, 3)  
)
```

```
# Install package if required
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Violin plot

ggplot(

  vehicle_data,

  aes(

    x = factor(Safety_Rating),

    y = Fuel_Efficiency

  )

) +

  geom_violin(

    fill = "skyblue",

    color = "black"

  ) +

  geom_jitter(

    width = 0.08,

    size = 3

  ) +

  labs(

    title = "Fuel Efficiency by Safety Rating",

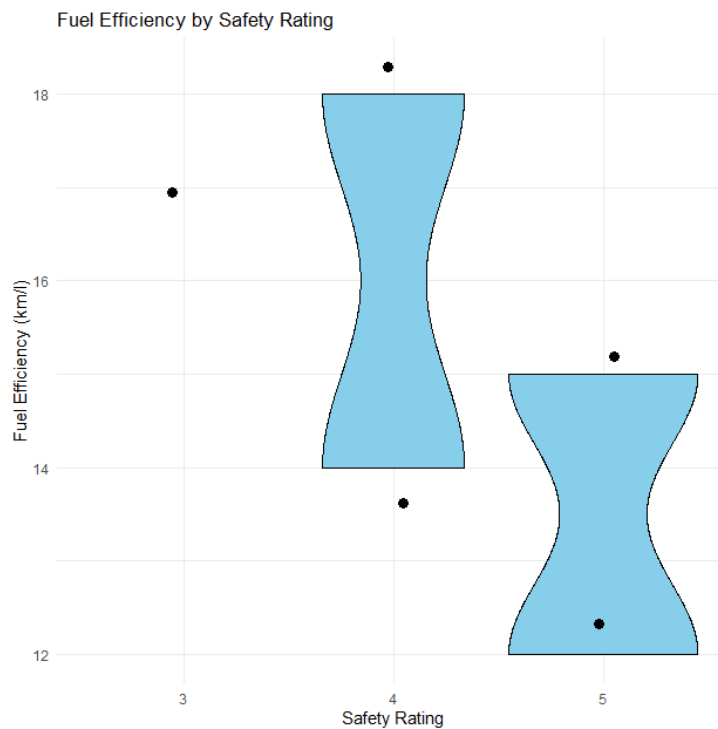
    x = "Safety Rating",

    y = "Fuel Efficiency (km/l)"
```

```
) +
```

```
theme_minimal()
```

Output:



- Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

R program:

```
library(ggplot2)
```

```
ggplot(
```

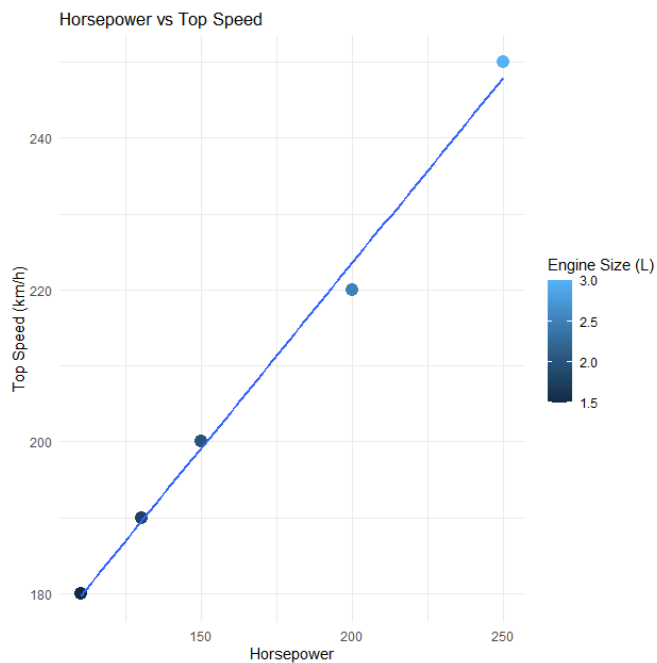
```
  vehicle_data,
```

```
  aes(
```

```
    x = Horsepower,
```

```
y = Top_Speed,  
  
color = Engine_Size  
  
)  
  
) +  
  
geom_point(  
  
size = 4  
  
) +  
  
geom_smooth(  
  
method = "lm",  
  
se = FALSE  
  
) +  
  
labs(  
  
title = "Horsepower vs Top Speed",  
  
x = "Horsepower",  
  
y = "Top Speed (km/h)",  
  
color = "Engine Size (L)"  
  
) +  
  
theme_minimal()
```

Output:



- Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

R program:

```
# Select numerical variables
```

```
numeric_data <- vehicle_data[
```

```
c(
```

```
"Engine_Size",
```

```
"Horsepower",
```

```
"Fuel_Efficiency",
```

```
"Top_Speed",
```

```
"Safety_Rating"
```

```
)
```

```
]
```

```
# Calculate correlation matrix
```

```
cor_matrix <- cor(numeric_data)
```

```
# Display correlation matrix
```

```
print(round(cor_matrix, 2))
```

```
# Create heatmap
```

```
heatmap(
```

```
  cor_matrix,
```

```
  Rowv = NA,
```

```
  Colv = NA,
```

```
  scale = "none",
```

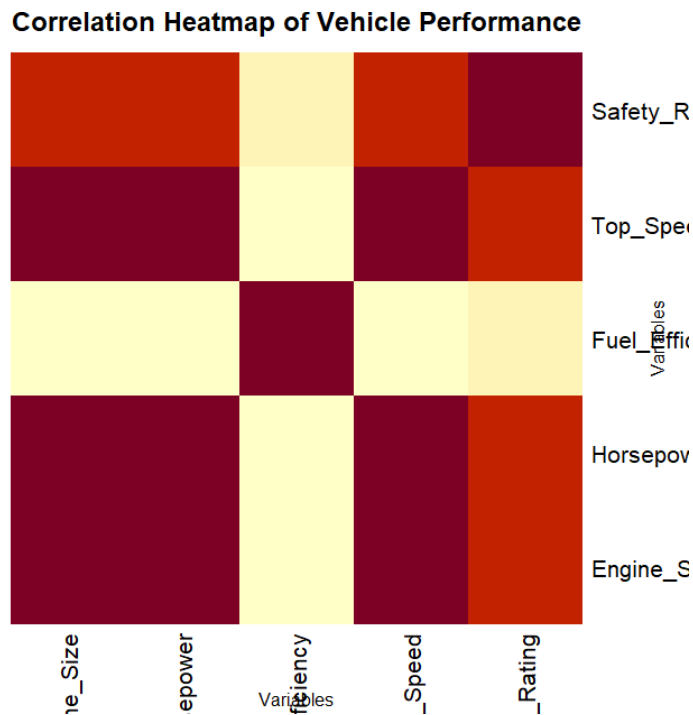
```
  main = "Correlation Heatmap of Vehicle Performance",
```

```
  xlab = "Variables",
```

```
  ylab = "Variables"
```

```
)
```


Output:



4. In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

Ans:

Step 1 – Import Data

Open Tableau:

Connect → Text File → Vehicle_Performance.csv

Step 2 – Bar Chart

Create a worksheet called:

Fuel Efficiency by Vehicle

Set:

Columns → Vehicle_ID

Rows → Fuel_Efficiency

Color → Safety_Rating

Select **Bar** from the Marks card.

Title

Fuel Efficiency by Vehicle and Safety Rating

Step 3 – Scatter Plot

Create another worksheet.

Set:

Columns → Horsepower

Rows → Top_Speed

Color → Engine_Size

Detail → Vehicle_ID

Select **Circle** from Marks.

Title

Horsepower vs Top Speed

This allows you to identify high-performance vehicles.

Step 4 – Add Engine Size Filter

Drag:

Engine_Size

to the **Filters** shelf.

Select:

Show Filter

Users can now select specific engine sizes.

Step 5 – Add Safety Rating Filter

Drag:

Safety_Rating

to the **Filters** shelf.

Select:

Show Filter

Users can explore vehicles according to safety rating.

Step 6 – Create Interactive Dashboard

Go to:

Dashboard → New Dashboard

Add:

1. **Fuel Efficiency by Vehicle**
2. **Horsepower vs Top Speed**
3. **Engine Size Filter**
4. **Safety Rating Filter**

Dashboard Layout

VEHICLE PERFORMANCE DASHBOARD	
Engine Size Filter	Safety Rating Filter
FUEL EFFICIENCY BY VEHICLE BAR CHART	
HORSEPOWER VS TOP SPEED SCATTER PLOT	

Dashboard Analysis

From the given data:

- **V3** has the highest performance: **250 HP** and **250 km/h**, but its fuel efficiency is only **12 km/l**.
- **V1** provides the highest fuel efficiency of **18 km/l**, with **110 HP** and **180 km/h**.
- **V2** provides a stronger balance between performance and safety, with **150 HP**, **200 km/h**, **15 km/l**, and a **5-star safety rating**.
- **V5** has good fuel efficiency (**17 km/l**) but a lower safety rating of **3**.
- There is a clear trade-off between **high performance and fuel efficiency**: higher horsepower and top speed are generally associated with lower fuel efficiency.

Conclusion

The dashboard helps compare vehicles based on **fuel efficiency, horsepower, top speed, engine size, and safety rating**. Among the five vehicles, **V2 can be considered a relatively balanced option**, while **V3 is the strongest performance-oriented vehicle** and **V1 is the most fuel-efficient**. These conclusions are based only on this very small sample and should not be generalized to all vehicles.

SET 29 .Student Academic Performance

Dataset:

Student_ID	Age	Study_Hours	Attendance (%)	Test_Score	Participation_Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

1. Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

R program:

```
# Create the dataset
```

```
student_data <- data.frame(
```

```
  Student_ID = c("S1", "S2", "S3", "S4", "S5"),
```

```
  Age = c(19, 21, 20, 22, 23),
```

```
  Study_Hours = c(12, 8, 15, 10, 7),
```

```
  Attendance = c(90, 70, 95, 85, 60),
```

```
  Test_Score = c(85, 70, 92, 80, 65),
```

```
  Participation_Score = c(8, 7, 9, 8, 6)
```

```
)
```

```
# Calculate cumulative values for stacked area
```

```
total_score <- student_data$Test_Score +
```

```
student_data$Participation_Score
```

```
# Create stacked area chart
```

```
plot(  
  
  1:nrow(student_data),  
  
  student_data$Test_Score,  
  
  type = "n",  
  
  xaxt = "n",  
  
  xlab = "Student",  
  
  ylab = "Score",  
  
  main = "Test Score and Participation Score"  
  
)
```

```
# Test Score area
```

```
polygon(  
  
  c(1:5, 5:1),  
  
  c(student_data$Test_Score, rep(0, 5)),  
  
  col = "skyblue",  
  
  border = "black"  
  
)
```

```
# Participation Score stacked on Test Score
```

```
polygon(
```

```
  c(1:5, 5:1),
```

```
  c(
```

```
    total_score,
```

```
    rev(student_data$Test_Score)
```

```
  ),
```

```
  col = "lightgreen",
```

```
  border = "black"
```

```
)
```

```
# Student labels
```

```
axis(
```

```
  1,
```

```
  at = 1:5,
```

```
  labels = student_data$Student_ID
```

```
)
```

```
# Legend
```

```

legend(

  "topleft",

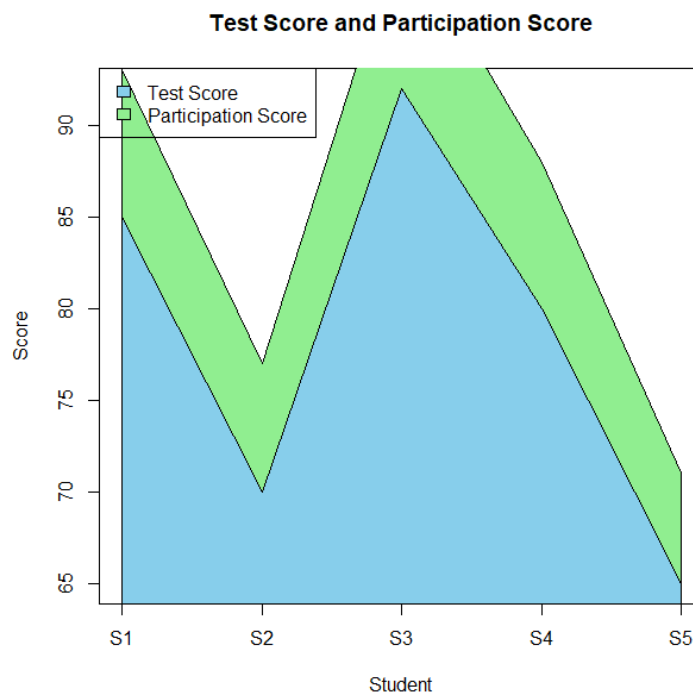
  legend = c("Test Score", "Participation Score"),

  fill = c("skyblue", "lightgreen")

)

```

Output:



- Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

Ans:

```
# Create attendance quartiles
```

```
student_data$Attendance_Quartile <- cut(
```

```
  student_data$Attendance,
```



```
breaks = quantile(

  student_data$Attendance,

  probs = c(0, 0.25, 0.50, 0.75, 1),

  na.rm = TRUE

),

include.lowest = TRUE,

labels = c(

  "Q1: Low",

  "Q2",

  "Q3",

  "Q4: High"

)

)

# Display the data

print(student_data)

# Create boxplot

boxplot(

  Study_Hours ~ Attendance_Quartile,
```

```

data = student_data,

main = "Study Hours by Attendance Quartiles",

xlab = "Attendance Quartile",

ylab = "Study Hours",

col = c(

  "lightblue",

  "lightgreen",

  "orange",

  "pink"

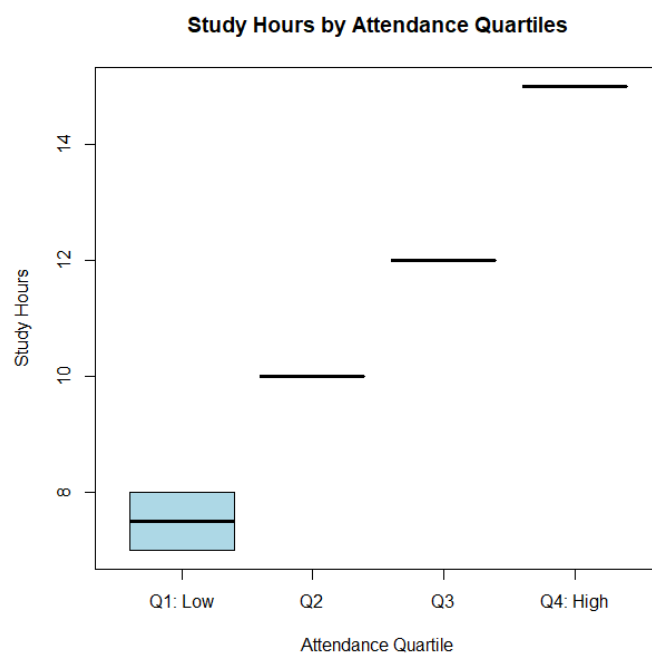
),

border = "black"

)

```

Output:



3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

R program:

```
# Create density plot
```

```
plot(
```

```
  density(student_data$Test_Score),
```

```
  main = "Density Plot of Test Scores",
```

```
  xlab = "Test Score",
```

```
  ylab = "Density",
```

```
  col = "blue",
```

```
  lwd = 2
```

```
)
```

```
# Add rug showing individual observations
```

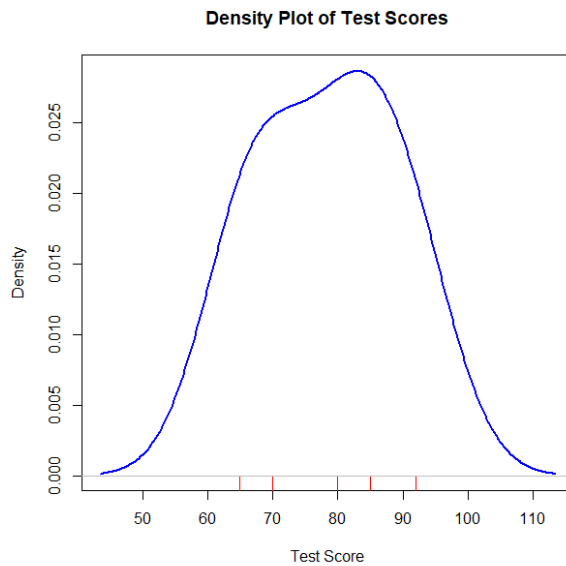
```
rug(
```

```
  student_data$Test_Score,
```

```
  col = "red"
```

```
)
```

Output:



4. In Tableau, design an interactive scatterplot with Study_Hours on X-axis, Test_Score on Y-axis, colored by Participation_Score. Include filters for Age.

Ans:

The objective is:

X-axis → Study_Hours

Y-axis → Test_Score

Color → Participation_Score

Filter → Age

Step 1: Import Dataset

1. Open **Tableau**.
2. Select **Connect** → **Text File**.
3. Select Student_Academic_Performance.csv.
4. Open **Sheet 1**.

Step 2: Create Scatter Plot

Drag:

- **Study_Hours** → Columns
- **Test_Score** → Rows
- **Participation_Score** → Color
- **Student_ID** → Detail

The result will be a scatter plot where each point represents one student.

Step 3: Make Participation Score a Color

In the **Marks** card:

1. Drag Participation_Score to **Color**.
2. Choose a suitable color palette.
3. Higher participation scores will have different colors from lower scores.

Step 4: Add Age Filter

1. Drag **Age** to **Filters**.
2. Select **All**.
3. Right-click Age.
4. Select **Show Filter**.

Now you can select different ages and the scatter plot will update.

Step 5: Add Student Information

Drag Student_ID to **Label** if you want the student names to appear on the points.

Your final Tableau setup should be:

Columns:

Study_Hours

Rows:

Test_Score

Marks:

Color → Participation_Score

Detail → Student_ID

Label → Student_ID

Filter:

Age

Tableau Interpretation

The scatter plot shows the relationship between study hours and test scores. In this dataset, students who study more hours generally achieve higher test scores. For example, **S3 studies 15 hours and scores 92**, while **S5 studies 7 hours and scores 65**. Participation score is represented using color, allowing students with different participation levels to be compared easily. The Age filter makes the visualization interactive and allows analysis of students of selected ages.

SET 30

User_ID	Gender	Age	Screen_Time (hrs)	App_Usage_Count	Data_Used (GB)	Satisfaction	Usage_Date
U01	Male	20	4.5	18	2.4	3	2025-01-08
U02	Female	22	6.0	25	3.8	5	2025-01-08
U03	Male	19	3.2	12	1.6	3	2025-02-11
U04	Female	21	7.1	30	4.5	5	2025-02-11
U05	Male	23	2.8	10	1.2	2	2025-03-14
U06	Female	20	5.4	22	3.1	4	2025-03-14

1. Import the dataset **in R**. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

Ans:

```
# Install ggplot2 if needed
```

```
install.packages("ggplot2")
```

```
# Load package
```

```
library(ggplot2)
```

```
# Create dataset directly in R
```

```
mobile <- data.frame(
```

```
  User_ID = c("U01", "U02", "U03", "U04", "U05", "U06"),
```

```
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
```

```
  Age = c(20, 22, 19, 21, 23, 20),
```

```
  Screen_Time = c(4.5, 6.0, 3.2, 7.1, 2.8, 5.4),
```

```
  App_Usage_Count = c(18, 25, 12, 30, 10, 22),
```

```
  Data_Used = c(2.4, 3.8, 1.6, 4.5, 1.2, 3.1),
```

```
  Satisfaction = c(3, 5, 3, 5, 2, 4)
```

```
)
```

```
# Histogram
```

```
ggplot(mobile, aes(x = Screen_Time)) +
```

```
  geom_histogram(
```

```
    bins = 6,
```

```
    fill = "skyblue",
```

```
    color = "black"
```

```
  ) +
```

```
  labs(
```

```
    title = "Distribution of Screen Time",
```

```
    x = "Screen Time (Hours)",
```

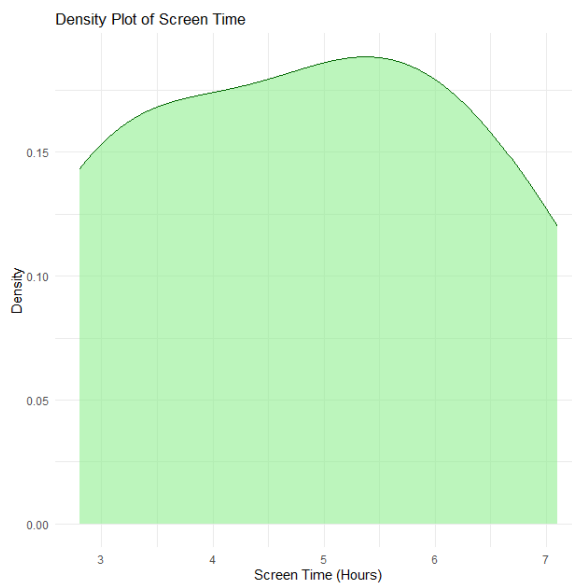
```
    y = "Number of Users"
```

```
  ) +
```

```
  theme_minimal()
```

```
# Density plot
ggplot(mobile, aes(x = Screen_Time)) +
  geom_density(
    fill = "lightgreen",
    color = "darkgreen",
    alpha = 0.6
  ) +
  labs(
    title = "Density Plot of Screen Time",
    x = "Screen Time (Hours)",
    y = "Density"
  ) +
  theme_minimal()
```

Output:



2. Create a scatter plot using **R programming** between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

Ans:

```
library(ggplot2)
```

```
# Create dataset
```

```
mobile <- data.frame(
  User_ID = c("U01", "U02", "U03", "U04", "U05", "U06"),
  Screen_Time = c(4.5, 6.0, 3.2, 7.1, 2.8, 5.4),
```



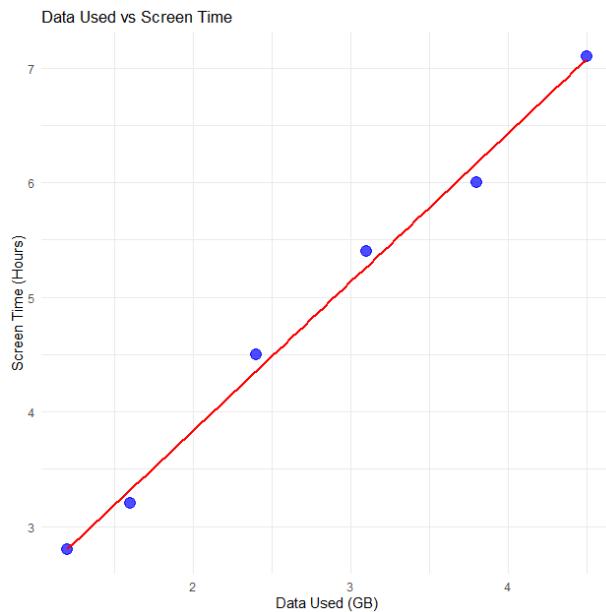
```
Data_Used = c(2.4, 3.8, 1.6, 4.5, 1.2, 3.1)
)
```

```
# Calculate correlation
correlation <- cor(
  mobile$Data_Used,
  mobile$Screen_Time
)
```

```
print(correlation)
```

```
# Scatter plot with trend line
ggplot(mobile, aes(
  x = Data_Used,
  y = Screen_Time
)) +
  geom_point(
    size = 4,
    color = "blue",
    alpha = 0.7
  ) +
  geom_smooth(
    method = "lm",
    se = FALSE,
    color = "red"
  ) +
  labs(
    title = "Data Used vs Screen Time",
    x = "Data Used (GB)",
    y = "Screen Time (Hours)"
  ) +
  theme_minimal()
```

Output:



3. **Using R**, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

Ans:

```
library(ggplot2)
```

```
# Create dataset
```

```
mobile <- data.frame(
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Satisfaction = c(3, 5, 3, 5, 2, 4)
)
```

```
# Calculate average satisfaction
```

```
average <- aggregate(
  Satisfaction ~ Gender,
  data = mobile,
  FUN = mean
)
```

```
# Display average values
```

```
print(average)
```

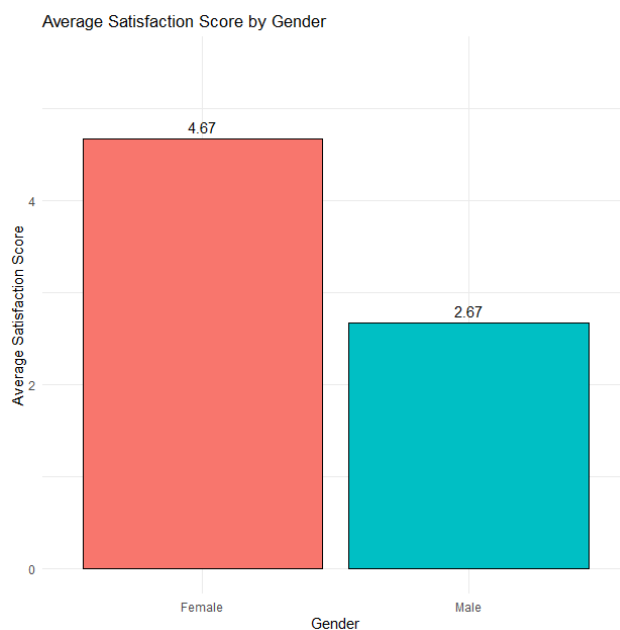
```
# Create bar chart
```

```

ggplot(average, aes(
  x = Gender,
  y = Satisfaction,
  fill = Gender
)) +
  geom_bar(
    stat = "identity",
    color = "black"
  ) +
  geom_text(
    aes(label = round(Satisfaction, 2)),
    vjust = -0.5
  ) +
  labs(
    title = "Average Satisfaction Score by Gender",
    x = "Gender",
    y = "Average Satisfaction Score"
  ) +
  ylim(0, 5.5) +
  theme_minimal() +
  theme(legend.position = "none")

```

Output:



4. Import Mobile_App_Usage.csv **into Tableau**. Create a bubble chart (Screen Time vs Data Used). Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage_Date. Design a mini dashboard.

Ans:

A. Create Bubble Chart

Requirements:

- X-axis → **Screen_Time**
- Y-axis → **Data_Used**
- Color → **Gender**
- Size → **App_Usage_Count**

Steps:

1. Open **Sheet 1**.
2. Drag **Screen_Time** → **Columns**.
3. Drag **Data_Used** → **Rows**.
4. In the **Marks** card, select **Circle**.
5. Drag **Gender** → **Color**.
6. Drag **App_Usage_Count** → **Size**.
7. Drag **User_ID** → **Detail**.

Rename the sheet:

Screen Time vs Data Used

B. Create Line Chart for Usage Trends

1. Click **New Worksheet**.
2. Drag **Usage_Date** → **Columns**.
3. Drag **Screen_Time** → **Rows**.
4. Change Marks to **Line**.
5. Drag **Gender** → **Color**.

Rename it:

Usage Trends

C. Add Gender Filter

- 1. Drag **Gender** → **Filters**.
- 2. Select **All**.
- 3. Right-click **Gender**.
- 4. Select **Show Filter**.

Now you can select **Male**, **Female**, or **All**.

D. Add Usage Date Filter

- 1. Drag **Usage_Date** → **Filters**.
- 2. Select **All**.
- 3. Right-click **Usage_Date**.
- 4. Select **Show Filter**.

Now you can select specific dates.

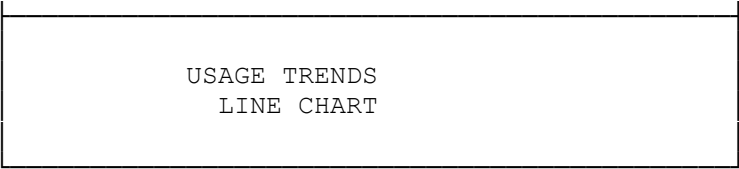
E. Create Mini Dashboard

Click:

New Dashboard

Add:

MOBILE APP USAGE DASHBOARD	
Gender Filter	Usage Date Filter
SCREEN TIME vs DATA USED BUBBLE CHART	



Final Interpretation

The bubble chart shows the relationship between **screen time and data usage**. Bubble size represents **App Usage Count**, while color represents **Gender**. The line chart shows how screen time changes across usage dates. The **Gender** and **Usage_Date** filters make the dashboard interactive and allow users to analyze specific groups and dates.